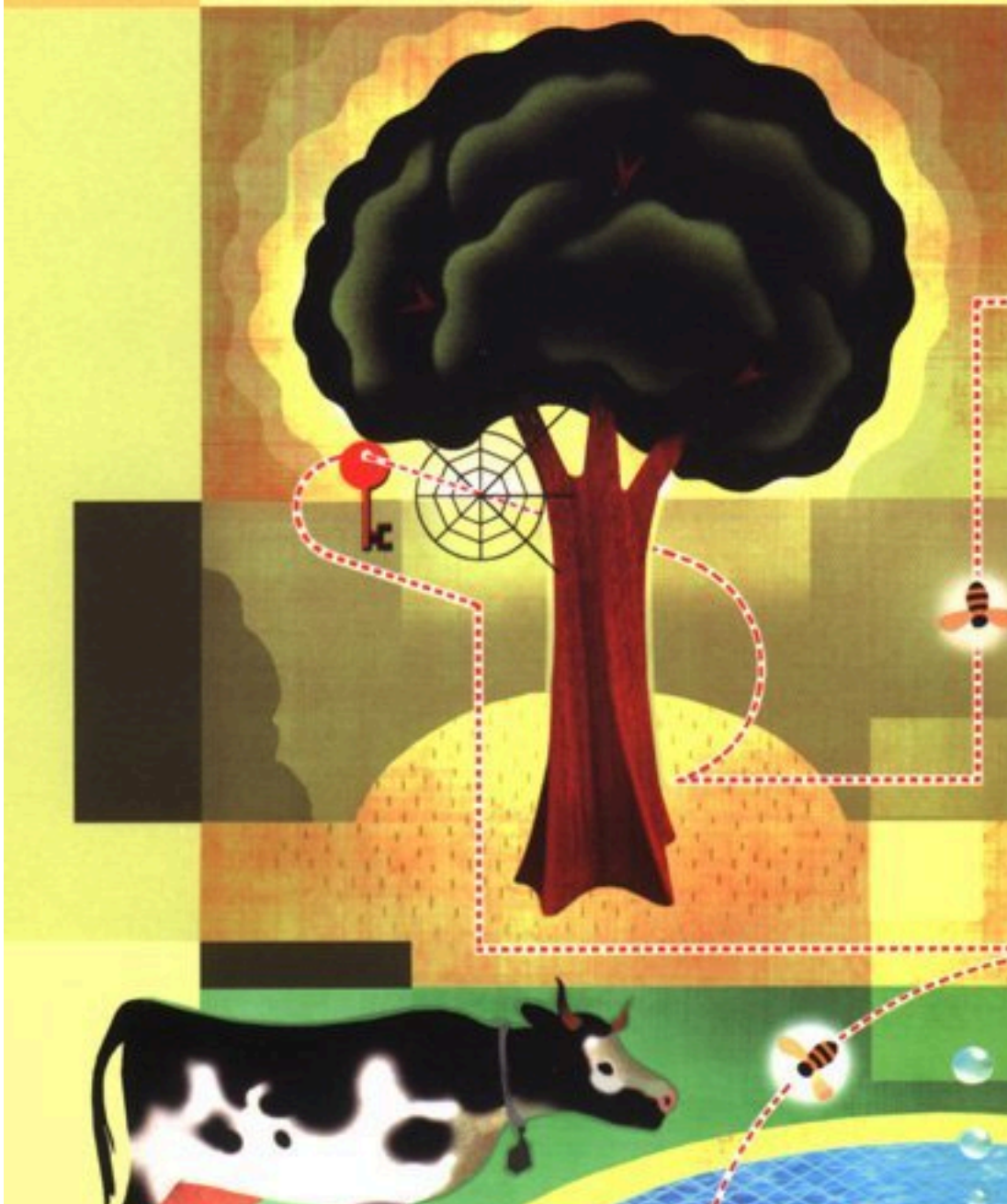
Database Manager

Second Editio



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PREFACE

The advantage of doing one's praising for oneself is that one can lay it on so thick and exactly in the right places.

—Samuel Butler

Database management systems have become ubiquitous as a fundamental tool for man aging information, and a course on the principles and practice of database systems is now an integral part of computer science curricula. This book covers the fundamentals of modern database management systems, in particular relational database systems. It is intended as a text for an introductory database course for undergraduates, and we have attempted to present the material in a clear, simple style.

A quantitative approach is used throughout and detailed examples abound. An exten sive set of exercises (for which solutions are available online to instructors) accompanies each chapter and reinforces students' ability to apply the concepts to real problems. The book contains enough material to support a second course, ideally supplemented by selected research papers. It can be used, with the accompanying software and SQL programming assignments, in two distinct kinds of introductory courses:

1. A course that aims to present the principles of database

systems, with a practical focus but without any implementation assignments. The SQL programming as signments are a useful supplement for such a course. The supplementary Minibase software can be used to create exercises and experiments with no programming.

2. A course that has a strong systems emphasis and assumes that students have good programming skills in C and C++. In this case the software can be used as the basis for projects in which students are asked to implement various parts of a relational DBMS. Several central modules in the project software (e.g., heap files, buffer manager, B+ trees, hash indexes, various join methods, concurrency control, and recovery algorithms) are described in sufficient detail in the text to enable students to implement them, given the (C++) class interfaces.

Many instructors will no doubt teach a course that falls

between these two extremes. xxii

Choice of Topics

The choice of material has been influenced by these considerations:

To concentrate on issues central to the *design, tuning, and implementation of rela tional database applications*. However, many of the issues discussed (e.g., buffering and access methods) are not specific to relational systems, and additional topics such as decision support and object-database systems are covered in later chapters.

To provide adequate coverage of implementation topics to support a concurrent laboratory section or course project. For example, implementation of relational operations has been covered in more detail than is necessary in a first course. However, the variety of alternative implementation techniques permits a wide choice of project assignments. An instructor who wishes to assign implementation of sort-merge join might cover that topic in depth, whereas another might choose to emphasize index nested loops join.

To provide in-depth coverage of the state of the art in currently available commer cial systems, rather than a broad coverage of several alternatives. For

example, we discuss the relational data model, B+ trees, SQL, System R style query op timization, lock-based concurrency control, the ARIES recovery algorithm, the two-phase commit protocol, asynchronous replication in distributed databases, and object-relational DBMSs in detail, with numerous illustrative examples. This is made possible by omitting or briefly covering some related topics such as the hierarchical and network models, B tree variants, Quel, semantic query optimiza tion, view serializability, the shadow-page recovery algorithm, and the three-phase commit protocol.

The same preference for in-depth coverage of selected topics governed our choice of topics for chapters on advanced material. Instead of covering a broad range of topics briefly, we have chosen topics that we believe to be practically important and at the cutting edge of current thinking in database systems, and we have covered them in depth.

New in the Second Edition

Based on extensive user surveys and feedback, we have refined the book's organization. The major change is the early introduction of the ER model, together with a discussion of conceptual database design. As in the first edition, we introduce SQL-92's data definition features together with the relational model (in Chapter 3), and whenever appropriate, relational model concepts (e.g., definition of a relation, updates, views, ER to relational mapping) are illustrated and discussed in the context of SQL. Of course, we maintain a careful separation between the concepts and their SQL realization. The material on data storage, file organization, and indexes has been moved back, and the

material on relational queries has been moved forward. Nonetheless, the two parts (storage and organization vs. queries) can still be taught in either order based on the instructor's preferences.

In order to facilitate brief coverage in a first course, the second edition contains overview chapters on transaction processing and query optimization. Most chapters have been revised extensively, and additional explanations and figures have been added in many places. For example, the chapters on query languages now contain a uniform numbering of all queries to facilitate comparisons of the same query (in algebra, calculus, and SQL), and the results of several queries are shown in figures. JDBC and ODBC coverage has been added to the SQL query chapter and SQL:1999 features are discussed both in this chapter and the chapter on object-relational databases. A discussion of RAID has been added to Chapter 7. We have added a new database design case study, illustrating the entire design cycle, as an appendix.

Two new pedagogical features have been introduced. First, 'floating boxes' provide ad ditional perspective and relate the concepts to real systems, while keeping the

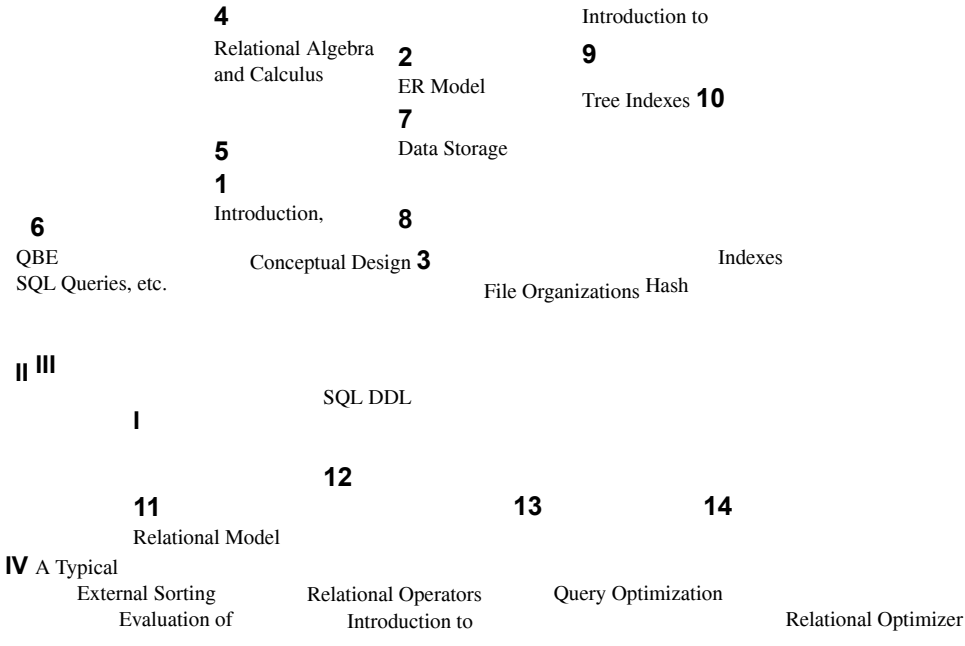
main dis cussion free of product-specific details. Second, each chapter concludes with a 'Points to Review' section that summarizes the main ideas introduced in the chapter and includes pointers to the sections where they are discussed.

For use in a second course, many advanced chapters from the first edition have been extended or split into multiple chapters to provide thorough coverage of current top ics. In particular, new material has been added to the chapters on decision support, deductive databases, and object databases. New chapters on Internet databases, data mining, and spatial databases have been added, greatly expanding the coverage of these topics.

The material can be divided into roughly seven parts, as indicated in Figure 0.1, which also shows the dependencies between chapters. An arrow from Chapter I to Chapter J means that I depends on material in J. The broken arrows indicate a weak dependency, which can be ignored at the instructor's discretion. It is recommended that Part I be covered first, followed by Part II and Part III (in either order). Other than these three parts, dependencies across parts are minimal.

Order of Presentation

The book's modular organization offers instructors a variety of choices. For exam ple, some instructors will want to cover SQL and get students to use a relational database, before discussing file organizations or indexing; they should cover Part II before Part III. In fact, in a course that emphasizes concepts and SQL, many of the implementation-oriented chapters might be skipped. On the other hand, instructors assigning implementation projects based on file organizations may want to cover Part



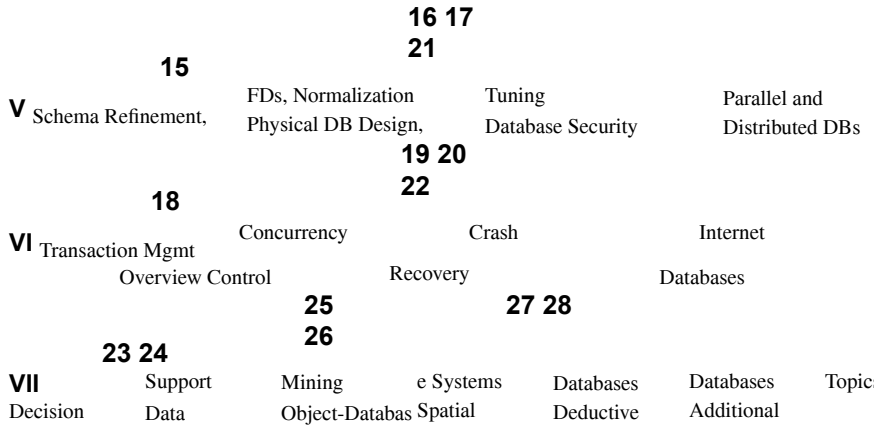


Figure 0.1 Chapter Organization and Dependencies

III early to space assignments. As another example, it is not necessary to cover all the alternatives for a given operator (e.g., various techniques for joins) in Chapter 12 in order to cover later related material (e.g., on optimization or tuning) adequately. The database design case study in the appendix can be discussed concurrently with the appropriate design chapters, or it can be discussed after all design topics have been covered, as a review.

Several section headings contain an asterisk. This symbol does not necessarily indicate a higher level of difficulty. Rather, omitting all asterisked sections leaves about the right amount of material in Chapters 1–18, possibly omitting Chapters 6, 10, and 14, for a broad introductory one-quarter or one-semester course (depending on the depth at which the remaining material is discussed and the nature of the course assignments).

The book can be used in several kinds of introductory or second courses by choosing topics appropriately, or in a two-course sequence by supplementing the material with some advanced readings in the second course. Examples of appropriate introductory courses include courses on file organizations and introduction to database management systems, especially if the course focuses on relational database design or implementa tion. Advanced courses can be built around the later chapters, which contain detailed bibliographies with ample pointers for further study.

Supplementary Material

Each chapter contains several exercises designed to test and expand the reader's un derstanding of the material. Students can obtain solutions to odd-numbered chapter exercises and a set of lecture slides for each chapter through the Web in Postscript and Adobe PDF formats.

The following material is available online to instructors:

1. Lecture slides for all chapters in MS Powerpoint, Postscript, and PDF formats.
2. Solutions to all chapter exercises.
3. SQL queries and programming assignments with solutions. (This is new for the second edition.)
4. Supplementary project software (Minibase) with sample assignments and solu tions, as described in Appendix B. The text itself does not refer to the project software, however, and can be used independently in a course that presents the principles of database management systems from a practical perspective, but with out a project component.

The supplementary material on SQL is new for the second edition. The remaining material has been extensively revised from the first edition versions.

For More Information

The home page for this book is at URL:

http://www.cs.wisc.edu/~dbbook

This page is frequently updated *and contains a link to all known errors in the book, the accompanying slides, and the supplements.* Instructors should visit this site periodically or register at this site to be notified of important changes by email.

Acknowledgments

This book grew out of lecture notes for CS564, the introductory (senior/graduate level) database course at UW-Madison. David DeWitt developed this course and the Minirel project, in which students wrote several well-chosen parts of a relational DBMS. My thinking about this material was shaped by teaching CS564, and Minirel was the inspiration for Minibase, which is more comprehensive (e.g., it has a query optimizer and includes visualization software) but tries to retain the spirit of Minirel. Mike Carey and I jointly designed much of Minibase. My lecture notes (and in turn this book) were influenced by Mike's lecture notes and by Yannis Ioannidis's lecture slides.

Joe Hellerstein used the beta edition of the book at Berkeley and provided invaluable feedback, assistance on slides, and hilarious quotes. Writing the chapter on object database systems with Joe was a lot of fun.

C. Mohan provided invaluable assistance, patiently answering a number of

questions about implementation techniques used in various commercial systems, in particular in dexing, concurrency control, and recovery algorithms. Moshe Zloof answered numerous questions about QBE semantics and commercial systems based on QBE. Ron Fagin, Krishna Kulkarni, Len Shapiro, Jim Melton, Dennis Shasha, and Dirk Van Gucht re viewed the book and provided detailed feedback, greatly improving the content and presentation. Michael Goldweber at Beloit College, Matthew Haines at Wyoming, Michael Kifer at SUNY StonyBrook, Jeff Naughton at Wisconsin, Praveen Seshadri at Cornell, and Stan Zdonik at Brown also used the beta edition in their database courses and offered feedback and bug reports. In particular, Michael Kifer pointed out an er ror in the (old) algorithm for computing a minimal cover and suggested covering some SQL features in Chapter 2 to improve modularity. Gio Wiederhold's bibliography, converted to Latex format by S. Sudarshan, and Michael Ley's online bibliography on databases and logic programming were a great help while compiling the chapter bibli ographies. Shaun Flisakowski and Uri Shaft helped me frequently in my never-ending battles with Latex.

I owe a special thanks to the many, many students who have contributed to the Mini base software. Emmanuel Ackaouy, Jim Pruyne, Lee Schumacher, and Michael Lee worked with me when I developed the first version of Minibase (much of which was subsequently discarded, but which influenced the next version). Emmanuel Ackaouy and Bryan So were my TAs when I taught CS564 using this version and went well be yond the limits of a TAship in their efforts to refine the project. Paul Aoki struggled with a version of Minibase and offered lots of useful comments as a TA at Berkeley. An entire class of CS764 students (our graduate database course) developed much of the current version of Minibase in a large class project that was led and coordinated by Mike Carey and me. Amit Shukla and Michael Lee were my TAs when I first taught CS564 using this version of Minibase and developed the software further.

Several students worked with me on independent projects, over a long period of time, to develop Minibase components. These include visualization packages for the buffer manager and B+ trees (Huseyin Bektas, Harry Stavropoulos, and Weiqing Huang); a query optimizer and visualizer (Stephen Harris, Michael Lee, and Donko Donjerkovic); an ER diagram tool based on the Opossum schema editor (Eben Haber); and a GUI based tool for normalization (Andrew Prock and Andy Therber). In addition, Bill Kimmel worked to integrate and fix a large body of code (storage manager, buffer manager, files and access methods, relational operators, and the query plan executor) produced by the CS764 class project. Ranjani Ramamurty considerably extended Bill's work on cleaning up and integrating the various modules. Luke Blanshard, Uri Shaft, and Shaun Flisakowski worked on putting together the release version of the code and developed test suites and exercises based on the Minibase software. Krishna Kunchithapadam tested the optimizer and developed part of the Minibase GUI.

Clearly, the Minibase software would not exist without the contributions of a great

many talented people. With this software available freely in the public domain, I hope that more instructors will be able to teach a systems-oriented database course with a blend of implementation and experimentation to complement the lecture material.

I'd like to thank the many students who helped in developing and checking the solu tions to the exercises and provided useful feedback on draft versions of the book. In alphabetical order: X. Bao, S. Biao, M. Chakrabarti, C. Chan, W. Chen, N. Cheung, D. Colwell, C. Fritz, V. Ganti, J. Gehrke, G. Glass, V. Gopalakrishnan, M. Higgins, T. Jasmin, M. Krishnaprasad, Y. Lin, C. Liu, M. Lusignan, H. Modi, S. Narayanan, D. Randolph, A. Ranganathan, J. Reminga, A. Therber, M. Thomas, Q. Wang, R. Wang, Z. Wang, and J. Yuan. Arcady Grenader, James Harrington, and Martin Reames at Wisconsin and Nina Tang at Berkeley provided especially detailed feedback.

Charlie Fischer, Avi Silberschatz, and Jeff Ullman gave me invaluable advice on work ing with a publisher. My editors at McGraw-Hill, Betsy Jones and Eric Munson, obtained extensive reviews and guided this book in its early stages. Emily Gray and Brad Kosirog were there whenever problems cropped up. At Wisconsin, Ginny Werner really helped me to stay on top of things.

Finally, this book was a thief of time, and in many ways it was harder on my family than on me. My sons expressed themselves forthrightly. From my (then) five-year old, Ketan: "Dad, stop working on that silly book. You don't have any time for *me*." Two-year-old Vivek: "You working *boook*? No no no come play basketball me!" All the seasons of their discontent were visited upon my wife, and Apu nonetheless cheerfully kept the family going in its usual chaotic, happy way all the many evenings and weekends I was wrapped up in this book. (Not to mention the days when I was wrapped up in being a faculty member!) As in all things, I can trace my parents' hand in much of this; my father, with his love of learning, and my mother, with her love of us, shaped me. My brother Kartik's contributions to this book consisted chiefly of

phone calls in which he kept me from working, but if I don't acknowledge him, he's liable to be annoyed. I'd like to thank my family for being there and giving meaning to everything I do. (There! I knew I'd find a legitimate reason to thank Kartik.)

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After reading about himself in the acknowledgment to the first edition, Ketan (now 8) had a simple question: “How come you didn’t dedicate the book to us? Why mom?” Ketan, I took care of this inexplicable oversight. Vivek (now 5) was more concerned about the extent of his fame: “Daddy, is my name in *evvy* copy of your book? Do they have it in *evvy* compooter science department in the world?” Vivek, I hope so. Finally, this revision would not have made it without Apu’s and Keiko’s support.

PART I BASICS

1 INTRODUCTION TO DATABASE SYSTEMS

Has everyone noticed that all the letters of the word *database* are typed with the left hand? Now the layout of the QWERTY typewriter keyboard was designed, among other things, to facilitate the even use of both hands. It follows, therefore, that

writing about databases is not only unnatural, but a lot harder than it appears.

—Anonymous

Today, more than at any previous time, the success of an organization depends on its ability to acquire accurate and timely data about its operations, to manage this data effectively, and to use it to analyze and guide its activities. Phrases such as the *information superhighway* have become ubiquitous, and information processing is a rapidly growing multibillion dollar industry.

The amount of information available to us is literally exploding, and the value of data as an organizational asset is widely recognized. Yet without the ability to manage this vast amount of data, and to quickly find the information that is relevant to a given question, as the amount of information increases, it tends to become a distraction and a liability, rather than an asset. This paradox drives the need for increasingly powerful and flexible data management systems. To get the most out of their large and complex datasets, users must have tools that simplify the tasks of managing the data and extracting useful information in a timely fashion. Otherwise, data can become a liability, with the cost of acquiring it and managing it far exceeding the value that is derived from it.

A database is a collection of data, typically describing the activities of one or more related organizations. For example, a university database might contain information about the following:

Entities such as students, faculty, courses, and classrooms.

Relationships between entities, such as students' enrollment in courses, faculty teaching courses, and the use of rooms for courses.

A database management system, or DBMS, is software designed to assist in maintaining and utilizing large collections of data, and the need for such systems, as well as their use, is growing rapidly. The alternative to using a DBMS is to use ad

hoc approaches that do not carry over from one application to another; for example, to store the data in files and write application-specific code to manage it. The use of a DBMS has several important advantages, as we will see in Section 1.4.

The area of database management systems is a microcosm of computer science in gen eral. The issues addressed and the techniques used span a wide spectrum, including languages, object-orientation and other programming paradigms, compilation, oper ating systems, concurrent programming, data structures, algorithms, theory, parallel and distributed systems, user interfaces, expert systems

and artificial intelligence, sta tistical techniques, and dynamic programming. We will not be able to go into all these aspects of database management in this book, but it should be clear that this is a rich and vibrant discipline.

1.1 OVERVIEW

The goal of this book is to present an in-depth introduction to database management systems, with an emphasis on how to organize information in a DBMS and to main tain it and retrieve it efficiently, that is, how to *design* a database and *use* a DBMS effectively. Not surprisingly, many decisions about how to use a DBMS for a given application depend on what capabilities the DBMS supports efficiently. Thus, to use a DBMS well, it is necessary to also understand how a DBMS *works*. The approach taken in this book is to emphasize how to *use* a DBMS, while covering DBMS implementation and architecture in sufficient detail to understand how to *design a database*.

Many kinds of database management systems are in use, but this book concentrates on *relational* systems, which are by far the dominant type of DBMS today. The following questions are addressed in the core chapters of this book:

1. Database Design: How can a user describe a real-world enterprise (e.g., a uni versity) in terms of the data stored in a DBMS? What factors must be considered in deciding how to organize the stored data? (Chapters 2, 3, 15, 16, and 17.)

2. Data Analysis: How can a user answer questions about the enterprise by posing queries over the data in the DBMS? (Chapters 4, 5, 6, and 23.)

3. Concurrency and Robustness: How does a DBMS allow many users to access data concurrently, and how does it protect the data in the event of system failures? (Chapters 18, 19, and 20.)

4. Efficiency and Scalability: How does a DBMS store large datasets and answer questions against this data efficiently? (Chapters 7, 8, 9, 10, 11, 12, 13, and 14.)

Later chapters cover important and rapidly evolving topics such as parallel and dis tributed database management, Internet databases, data warehousing and complex

queries for decision support, data mining, object databases, spatial data management, and rule-oriented DBMS extensions.

In the rest of this chapter, we introduce the issues listed above. In Section 1.2, we begin with a brief history of the field and a discussion of the role of database management in modern information systems. We then identify benefits of storing data in a DBMS instead of a file system in Section 1.3, and discuss the advantages of using a DBMS to manage data in Section 1.4. In Section 1.5 we consider how

information about an enterprise should be organized and stored in a DBMS. A user probably thinks about this information in high-level terms corresponding to the entities in the organization and their relationships, whereas the DBMS ultimately stores data in the form of (many, many) bits. The gap between how users think of their data and how the data is ultimately stored is bridged through several *levels of abstraction* supported by the DBMS. Intuitively, a user can begin by describing the data in fairly high-level terms, and then refine this description by considering additional storage and representation details as needed.

In Section 1.6 we consider how users can retrieve data stored in a DBMS and the need for techniques to efficiently compute answers to questions involving such data. In Section 1.7 we provide an overview of how a DBMS supports concurrent access to data by several users, and how it protects the data in the event of system failures.

We then briefly describe the internal structure of a DBMS in Section 1.8, and mention various groups of people associated with the development and use of a DBMS in Section 1.9.

1.2 A HISTORICAL PERSPECTIVE

From the earliest days of computers, storing and manipulating data have been a major application focus. The first general-purpose DBMS was designed by Charles Bachman at General Electric in the early 1960s and was called the Integrated Data Store. It formed the basis for the *network data model*, which was standardized by the Conference on Data Systems Languages (CODASYL) and strongly influenced database systems through the 1960s. Bachman was the first recipient of ACM's Turing Award (the computer science equivalent of a Nobel prize) for work in the database area; he received the award in 1973.

In the late 1960s, IBM developed the Information Management System (IMS) DBMS, used even today in many major installations. IMS formed the basis for an alternative data representation framework called the *hierarchical data model*. The SABRE system for making airline reservations was jointly developed by American Airlines and IBM around the same time, and it allowed several people to access the same data through

a computer network. Interestingly, today the same SABRE system is used to power popular Web-based travel services such as Travelocity!

In 1970, Edgar Codd, at IBM's San Jose Research Laboratory, proposed a new data representation framework called the *relational data model*. This proved to be a water shed in the development of database systems: it sparked rapid development of several DBMSs based on the relational model, along with a rich body of theoretical results that placed the field on a firm foundation. Codd won the 1981 Turing Award for his seminal work. Database systems matured as an academic discipline, and the

popu larity of relational DBMSs changed the commercial landscape. Their benefits were widely recognized, and the use of DBMSs for managing corporate data became stan dard practice.

In the 1980s, the relational model consolidated its position as the dominant DBMS paradigm, and database systems continued to gain widespread use. The SQL query language for relational databases, developed as part of IBM's System R project, is now the standard query language. SQL was standardized in the late 1980s, and the current standard, SQL-92, was adopted by the American National Standards Institute (ANSI) and International Standards Organization (ISO). Arguably, the most widely used form of concurrent programming is the concurrent execution of database programs (called *transactions*). Users write programs as if they are to be run by themselves, and the responsibility for running them concurrently is given to the DBMS. James Gray won the 1999 Turing award for his contributions to the field of transaction management in a DBMS.

In the late 1980s and the 1990s, advances have been made in many areas of database systems. Considerable research has been carried out into more powerful query lan guages and richer data models, and there has been a big emphasis on supporting complex analysis of data from all parts of an enterprise. Several vendors (e.g., IBM's DB2, Oracle 8, Informix UDS) have extended their systems with the ability to store new data types such as images and text, and with the ability to ask more complex queries. Specialized systems have been developed by numerous vendors for creating *data warehouses*, consolidating data from several databases, and for carrying out spe cialized analysis.

An interesting phenomenon is the emergence of several *enterprise resource planning (ERP)* and *management resource planning (MRP)* packages, which add a substantial layer of application-oriented features on top of a DBMS. Widely used packages include systems from Baan, Oracle, PeopleSoft, SAP, and Siebel. These packages identify a set of common tasks (e.g., inventory management, human resources planning, finan cial analysis) encountered by a large number of organizations and provide a general application layer to carry out these tasks. The data is stored in a relational DBMS, and the application layer can be customized to different companies, leading to lower

overall costs for the companies, compared to the cost of building the application layer from scratch.

Most significantly, perhaps, DBMSs have entered the Internet Age. While the first generation of Web sites stored their data exclusively in operating systems files, the use of a DBMS to store data that is accessed through a Web browser is becoming widespread. Queries are generated through Web-accessible forms and answers are formatted using a markup language such as HTML, in order to be easily displayed in a browser. All the database vendors are adding features to their DBMS aimed at making it more suitable for deployment over the Internet.

Database management continues to gain importance as more and more data is brought on-line, and made ever more accessible through computer networking. Today the field is being driven by exciting visions such as multimedia databases, interactive video, digital libraries, a host of scientific projects such as the human genome mapping effort and NASA's Earth Observation System project, and the desire of companies to consolidate their decision-making processes and *mine* their data repositories for useful information about their businesses. Commercially, database management systems represent one of the largest and most vigorous market segments. Thus the study of database systems could prove to be richly rewarding in more ways than one!

1.3 FILE SYSTEMS VERSUS A DBMS

To understand the need for a DBMS, let us consider a motivating scenario: A company has a large collection (say, 500 GB[1]) of data on employees, departments, products, sales, and so on. This data is accessed concurrently by several employees. Questions about the data must be answered quickly, changes made to the data by different users must be applied consistently, and access to certain parts of the data (e.g., salaries) must be restricted.

We can try to deal with this data management problem by storing the data in a collection of operating system files. This approach has many drawbacks, including the following:

We probably do not have 500 GB of main memory to hold all the data. We must therefore store data in a storage device such as a disk or tape and bring relevant parts into main memory for processing as needed.

Even if we have 500 GB of main memory, on computer systems with 32-bit ad dressing, we cannot refer directly to more than about 4 GB of data! We have to program some method of identifying all data items.

[1]A kilobyte (KB) is 1024 bytes, a megabyte (MB) is 1024 KBs, a gigabyte (GB) is 1024 MBs, a terabyte (TB) is 1024 GBs, and a petabyte (PB) is 1024 terabytes.

We have to write special programs to answer each question that users may want to ask about the data. These programs are likely to be complex because of the large volume of data to be searched.

We must protect the data from inconsistent changes made by different users ac cessing the data concurrently. If programs that access the data are written with such concurrent access in mind, this adds greatly to their complexity.

We must ensure that data is restored to a consistent state if the system crashes while changes are being made.

Operating systems provide only a password mechanism for security. This is not sufficiently flexible to enforce security policies in which different users have per mission to access different subsets of the data.

A DBMS is a piece of software that is designed to make the preceding tasks easier. By storing data in a DBMS, rather than as a collection of operating system files, we can use the DBMS's features to manage the data in a robust and efficient manner. As the volume of data and the number of users grow—hundreds of gigabytes of data and thousands of users are common in current corporate databases—DBMS support becomes indispensable.

1.4 ADVANTAGES OF A DBMS

Using a DBMS to manage data has many advantages:

Data independence: Application programs should be as independent as possi ble from details of data representation and storage. The DBMS can provide an abstract view of the data to insulate application code from such details.

Efficient data access: A DBMS utilizes a variety of sophisticated techniques to store and retrieve data efficiently. This feature is especially important if the data is stored on external storage devices.

Data integrity and security: If data is always accessed through the DBMS, the DBMS can enforce integrity constraints on the data. For example, before inserting salary information for an employee, the DBMS can check that the department budget is not exceeded. Also, the DBMS can enforce *access controls* that govern what data is visible to different classes of users.

Data administration: When several users share the data, centralizing the ad ministration of data can offer significant improvements. Experienced professionals who understand the nature of the data being managed, and how different groups of users use it, can be responsible for organizing the data representation to min imize redundancy and for fine-tuning the storage of the data to make retrieval efficient.

Concurrent access and crash recovery: A DBMS schedules concurrent ac cesses to the data in such a manner that users can think of the data as being accessed by only one user at a time. Further, the DBMS protects users from the effects of system failures.

Reduced application development time: Clearly, the DBMS supports many important functions that are common to many applications accessing data stored in the DBMS. This, in conjunction with the high-level interface to the data, facil itates quick development of applications. Such applications are also likely to be more robust than applications developed from scratch because many

important tasks are handled by the DBMS instead of being implemented by the application.

Given all these advantages, is there ever a reason *not* to use a DBMS? A DBMS is a complex piece of software, optimized for certain kinds of workloads (e.g., answering complex queries or handling many concurrent requests), and its performance may not be adequate for certain specialized applications. Examples include applications with tight real-time constraints or applications with just a few well-defined critical opera tions for which efficient custom code must be written. Another reason for not using a DBMS is that an application may need to manipulate the data in ways not supported by the query language. In such a situation, the abstract view of the data presented by the DBMS does not match the application's needs, and actually gets in the way. As an example, relational databases do not support flexible analysis of text data (although vendors are now extending their products in this direction). If specialized performance or data manipulation requirements are central to an application, the application may choose not to use a DBMS, especially if the added benefits of a DBMS (e.g., flexible querying, security, concurrent access, and crash recovery) are not required. In most situations calling for large-scale data management, however, DBMSs have become an indispensable tool.

1.5 DESCRIBING AND STORING DATA IN A DBMS

The user of a DBMS is ultimately concerned with some real-world enterprise, and the data to be stored describes various aspects of this enterprise. For example, there are students, faculty, and courses in a university, and the data in a university database describes these entities and their relationships.

A data model is a collection of high-level data description constructs that hide many low-level storage details. A DBMS allows a user to define the data to be stored in terms of a data model. Most database management systems today are based on the relational data model, which we will focus on in this book.

While the data model of the DBMS hides many details, it is nonetheless closer to how the DBMS stores data than to how a user thinks about the underlying enterprise. A semantic data model is a more abstract, high-level data model that makes it easier

for a user to come up with a good initial description of the data in an enterprise. These models contain a wide variety of constructs that help describe a real application scenario. A DBMS is not intended to support all these constructs directly; it is typically built around a data model with just a few basic constructs, such as the relational model. A database design in terms of a semantic model serves as a useful starting point and is subsequently translated into a database design in terms of the data model the DBMS actually supports.

A widely used semantic data model called the entity-relationship (ER) model allows us to pictorially denote entities and the relationships among them. We cover the ER model in Chapter 2.

1.5.1 The Relational Model

In this section we provide a brief introduction to the relational model. The central data description construct in this model is a relation, which can be thought of as a set of records.

A description of data in terms of a data model is called a schema. In the relational model, the schema for a relation specifies its name, the name of each field (or attribute or column), and the type of each field. As an example, student information in a university database may be stored in a relation with the following schema:

Students(*sid:* string, *name:* string, *login:* string, *age:* integer, *gpa:* real)

The preceding schema says that each record in the Students relation has five fields, with field names and types as indicated.[2] An example instance of the Students relation appears in Figure 1.1.

sid	*name*	*login*	*age*	*gpa*
53666	Jones	jones@cs	18	3.4
53688	Smith	smith@ee	18	3.2
53650	Smith	smith@math	19	3.8
53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

Figure 1.1 An Instance of the Students Relation

[2]Storing *date of birth* is preferable to storing *age*, since it does not change over time, unlike age. We've used *age* for simplicity in our discussion.

Each row in the Students relation is a record that describes a student. The

description is not complete—for example, the student's height is not included—but is presumably adequate for the intended applications in the university database. Every row follows the schema of the Students relation. The schema can therefore be regarded as a template for describing a student.

We can make the description of a collection of students more precise by specifying integrity constraints, which are conditions that the records in a relation must satisfy. For example, we could specify that every student has a unique *sid* value. Observe that we cannot capture this information by simply adding another field to the Students schema. Thus, the ability to specify uniqueness of the values in a field increases the accuracy with which we can describe our data. The expressiveness of the constructs available for specifying integrity constraints is an important aspect of a data model.

Other Data Models

In addition to the relational data model (which is used in numerous systems, including IBM's DB2, Informix, Oracle, Sybase, Microsoft's Access, FoxBase, Paradox, Tandem, and Teradata), other important data models include the hierarchical model (e.g., used in IBM's IMS DBMS), the network model (e.g., used in IDS and IDMS), the object oriented model (e.g., used in Objectstore and Versant), and the object-relational model (e.g., used in DBMS products from IBM, Informix, ObjectStore, Oracle, Versant, and others). While there are many databases that use the hierarchical and network models, and systems based on the object-oriented and object-relational models are gaining acceptance in the marketplace, the dominant model today is the relational model.

In this book, we will focus on the relational model because of its wide use and impor tance. Indeed, the object-relational model, which is gaining in popularity, is an effort to combine the best features of the relational and object-oriented models, and a good grasp of the relational model is necessary to understand object-relational concepts. (We discuss the object-oriented and object-relational models in Chapter 25.)

1.5.2 Levels of Abstraction in a DBMS

The data in a DBMS is described at three levels of abstraction, as illustrated in Figure 1.2. The database description consists of a schema at each of these three levels of abstraction: the *conceptual*, *physical*, and *external* schemas.

A data definition language (DDL) is used to define the external and conceptual schemas. We will discuss the DDL facilities of the most widely used database language, SQL, in Chapter 3. All DBMS vendors also support SQL commands to describe aspects of the physical schema, but these commands are not part of the SQL-92 language

 External Schema 1 External Schema 2 External Schema 3

Conceptual Schema

Physical Schema

DISK

Figure 1.2 Levels of Abstraction in a DBMS

standard. Information about the conceptual, external, and physical schemas is stored in the system catalogs (Section 13.2). We discuss the three levels of abstraction in the rest of this section.

Conceptual Schema

The conceptual schema (sometimes called the logical schema) describes the stored data in terms of the data model of the DBMS. In a relational DBMS, the conceptual schema describes all relations that are stored in the database. In our sample university database, these relations contain information about *entities*, such as students and faculty, and about *relationships*, such as students' enrollment in courses. All student entities can be described using records in a Students relation, as we saw earlier. In fact, each collection of entities and each collection of relationships can be described as a relation, leading to the following conceptual schema:

Students(*sid:* string, *name:* string, *login:* string,
age: integer, *gpa:* real)
Faculty(*fid:* string, *fname:* string, *sal:* real)
Courses(*cid:* string, *cname:* string, *credits:* integer)
Rooms(*rno:* integer, *address:* string, *capacity:* integer)
Enrolled(*sid:* string, *cid:* string, *grade:* string)
Teaches(*fid:* string, *cid:* string)
Meets In(*cid:* string, *rno:* integer, *time:* string)

The choice of relations, and the choice of fields for each relation, is not always obvi ous, and the process of arriving at a good conceptual schema is called conceptual database design. We discuss conceptual database design in Chapters 2 and 15.

Physical Schema

The physical schema specifies additional storage details. Essentially, the physical schema summarizes how the relations described in the conceptual schema are actually stored on secondary storage devices such as disks and tapes.

We must decide what file organizations to use to store the relations, and create auxiliary data structures called indexes to speed up data retrieval operations. A sample physical schema for the university database follows:

Store all relations as unsorted files of records. (A file in a DBMS is either a collection of records or a collection of pages, rather than a string of characters as in an operating system.)

Create indexes on the first column of the Students, Faculty, and Courses relations, the *sal* column of Faculty, and the *capacity* column of Rooms.

Decisions about the physical schema are based on an understanding of how the data is typically accessed. The process of arriving at a good physical schema is called physical database design. We discuss physical database design in Chapter 16.

External Schema

External schemas, which usually are also in terms of the data model of the DBMS, allow data access to be customized (and authorized) at the level of individual users or groups of users. Any given database has exactly one conceptual schema and one physical schema because it has just one set of stored relations, but it may have several external schemas, each tailored to a particular group of users. Each external schema consists of a collection of one or more views and relations from the conceptual schema. A view is conceptually a relation, but the records in a view are not stored in the DBMS. Rather, they are computed using a definition for the view, in terms of relations stored in the DBMS. We discuss views in more detail in Chapter 3.

The external schema design is guided by end user requirements. For example, we might want to allow students to find out the names of faculty members teaching courses, as well as course enrollments. This can be done by defining the following view:

Courseinfo(*cid:* string, *fname:* string, *enrollment:* integer)

A user can treat a view just like a relation and ask questions about the records in the view. Even though the records in the view are not stored explicitly, they are computed as needed. We did not include Courseinfo in the conceptual schema because we can compute Courseinfo from the relations in the conceptual schema, and to store it in addition would be redundant. Such redundancy, in addition to the wasted space, could

lead to inconsistencies. For example, a tuple may be inserted into the Enrolled relation, indicating that a particular student has enrolled in some course, without incrementing the value in the *enrollment* field of the corresponding record of Courseinfo (if the latter also is part of the conceptual schema and its tuples are stored in the DBMS).

1.5.3 Data Independence

A very important advantage of using a DBMS is that it offers data independence. That is, application programs are insulated from changes in the way the data is struc tured and stored. Data independence is achieved through use of the three levels of data abstraction; in particular, the conceptual schema and the external schema pro vide distinct benefits in this area.

Relations in the external schema (view relations) are in principle generated on demand from the relations corresponding to the conceptual schema.[3] If the underlying data is reorganized, that is, the conceptual schema is changed, the definition of a view relation can be modified so that the same relation is computed as before. For example, suppose that the Faculty relation in our university database is replaced by the following two relations:

Faculty public(*fid:* string, *fname:* string, *office:* integer)
Faculty private(*fid:* string, *sal:* real)

Intuitively, some confidential information about faculty has been placed in a separate relation and information about offices has been added. The Courseinfo view relation can be redefined in terms of Faculty public and Faculty private, which together contain all the information in Faculty, so that a user who queries Courseinfo will get the same answers as before.

Thus users can be shielded from changes in the logical structure of the data, or changes in the choice of relations to be stored. This property is called logical data indepen dence.

In turn, the conceptual schema insulates users from changes in the physical storage of the data. This property is referred to as physical data independence. The conceptual schema hides details such as how the data is actually laid out on disk, the file structure, and the choice of indexes. As long as the conceptual schema remains the same, we can change these storage details without altering applications. (Of course, performance might be affected by such changes.)

[3]In practice, they could be precomputed and stored to speed up queries on view relations, but the computed view relations must be updated whenever the underlying relations are updated.

1.6 QUERIES IN A DBMS

The ease with which information can be obtained from a database often determines its value to a user. In contrast to older database systems, relational database systems allow a rich class of questions to be posed easily; this feature has contributed greatly to their popularity. Consider the sample university database in Section 1.5.2. Here are examples of questions that a user might ask:

1. What is the name of the student with student id 123456?
2. What is the average salary of professors who teach the course with cid

CS564? 3. How many students are enrolled in course CS564?

4. What fraction of students in course CS564 received a grade better than

B? 5. Is any student with a GPA less than 3.0 enrolled in course CS564?

Such questions involving the data stored in a DBMS are called queries. A DBMS provides a specialized language, called the query language, in which queries can be posed. A very attractive feature of the relational model is that it supports powerful query languages. Relational calculus is a formal query language based on mathemat ical logic, and queries in this language have an intuitive, precise meaning. Relational algebra is another formal query language, based on a collection of operators for manipulating relations, which is equivalent in power to the calculus.

A DBMS takes great care to evaluate queries as efficiently as possible. We discuss query optimization and evaluation in Chapters 12 and 13. Of course, the efficiency of query evaluation is determined to a large extent by how the data is stored physically. Indexes can be used to speed up many queries—in fact, a good choice of indexes for the underlying relations can speed up each query in the preceding list. We discuss data storage and indexing in Chapters 7, 8, 9, and 10.

A DBMS enables users to create, modify, and query data through a data manipula tion language (DML). Thus, the query language is only one part of the DML, which also provides constructs to insert, delete, and modify data. We will discuss the DML features of SQL in Chapter 5. The DML and DDL are collectively referred to as the data sublanguage when embedded within a host language (e.g., C or COBOL).

1.7 TRANSACTION MANAGEMENT

Consider a database that holds information about airline reservations. At any given instant, it is possible (and likely) that several travel agents are looking up information about available seats on various flights and making new seat reservations. When several users access (and possibly modify) a database concurrently, the DBMS must order

their requests carefully to avoid conflicts. For example, when one travel agent looks up Flight 100 on some given day and finds an empty seat, another travel agent may simultaneously be making a reservation for that seat, thereby making the information seen by the first agent obsolete.

Another example of concurrent use is a bank's database. While one user's application program is computing the total deposits, another application may transfer money from an account that the first application has just 'seen' to an account that has not yet been seen, thereby causing the total to appear larger than it should be. Clearly, such anomalies should not be allowed to occur. However, disallowing concurrent access can degrade performance.

Further, the DBMS must protect users from the effects of system failures by ensuring that all data (and the status of active applications) is restored to a consistent state when the system is restarted after a crash. For example, if a travel agent asks for a reservation to be made, and the DBMS responds saying that the reservation has been made, the reservation should not be lost if the system crashes. On the other hand, if the DBMS has not yet responded to the request, but is in the process of making the necessary changes to the data while the crash occurs, the partial changes should be undone when the system comes back up.

A transaction is *any one execution* of a user program in a DBMS. (Executing the same program several times will generate several transactions.) This is the basic unit of change as seen by the DBMS: Partial transactions are not allowed, and the effect of a group of transactions is equivalent to some serial execution of all transactions. We briefly outline how these properties are guaranteed, deferring a detailed discussion to later chapters.

1.7.1 Concurrent Execution of Transactions

An important task of a DBMS is to schedule concurrent accesses to data so that each user can safely ignore the fact that others are accessing the data concurrently. The im portance of this task cannot be underestimated because a database is typically shared by a large number of users, who submit their requests to the DBMS independently, and simply cannot be expected to deal with arbitrary changes being made concurrently by other users. A DBMS allows users to think of their programs as if they were executing in isolation, one after the other in some order chosen by the DBMS. For example, if a program that deposits cash into an account is submitted to the DBMS at the same time as another program that debits money from the same account, either of these programs could be run first by the DBMS, but their steps will not be interleaved in such a way that they interfere with each other.

A locking protocol is a set of rules to be followed by each transaction (and enforced

by the DBMS), in order to ensure that even though actions of several transactions might be interleaved, the net effect is identical to executing all transactions in some serial order. A lock is a mechanism used to control access to database objects. Two kinds of locks are commonly supported by a DBMS: shared locks on an object can be held by two different transactions at the same time, but an exclusive lock on an object ensures that no other transactions hold *any* lock on this object.

Suppose that the following locking protocol is followed: Every transaction begins by obtaining a shared lock on each data object that it needs to read and an exclusive lock on each data object that it needs to modify, and then releases all its locks after completing all actions. Consider two transactions $T1$ and $T2$ such that $T1$ wants to modify a data object and $T2$ wants to read the same object. Intuitively, if $T1$'s request for an exclusive lock on the object is granted first, $T2$ cannot proceed until $T1$ releases this lock, because $T2$'s request for a shared lock will not be granted by the DBMS until then. Thus, all of $T1$'s actions will be completed before any of $T2$'s actions are initiated. We consider locking in more detail in Chapters 18 and 19.

1.7.2 Incomplete Transactions and System Crashes

Transactions can be interrupted before running to completion for a variety of reasons, e.g., a system crash. A DBMS must ensure that the changes made by such incomplete transactions are removed from the database. For example, if the DBMS is in the middle of transferring money from account A to account B, and has debited the first account but not yet credited the second when the crash occurs, the money debited from account A must be restored when the system comes back up after the crash.

To do so, the DBMS maintains a log of all writes to the database. A crucial prop erty of the log is that each write action must be recorded in the log (on disk) *before* the corresponding change is reflected in the database itself—otherwise, if the system crashes just after making the change in the database but before the change is recorded in the log, the DBMS would be unable to detect and undo this change. This property is called Write-Ahead Log or WAL. To ensure this property, the DBMS must be able to selectively force a page in memory to disk.

The log is also used to ensure that the changes made by a successfully completed transaction are not lost due to a system crash, as explained in Chapter 20. Bringing the database to a consistent state after a system crash can be a slow process, since the DBMS must ensure that the effects of all transactions that completed prior to the crash are restored, and that the effects of incomplete transactions are undone. The time required to recover from a crash can be reduced by periodically forcing some information to disk; this periodic operation is called a checkpoint.

1.7.3 Points to Note

In summary, there are three points to remember with respect to DBMS support for concurrency control and recovery:

1. Every object that is read or written by a transaction is first locked in shared or exclusive mode, respectively. Placing a lock on an object restricts its availability to other transactions and thereby affects performance.

2. For efficient log maintenance, the DBMS must be able to selectively force a collec tion of pages in main memory to disk. Operating system support for this operation is not always satisfactory.

3. Periodic checkpointing can reduce the time needed to recover from a crash. Of course, this must be balanced against the fact that checkpointing too often slows down normal execution.

1.8 STRUCTURE OF A DBMS

Figure 1.3 shows the structure (with some simplification) of a typical DBMS based on the relational data model.

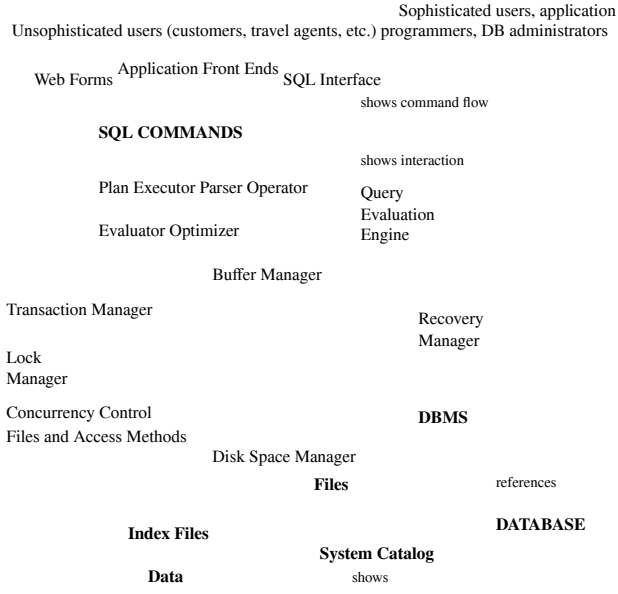

Figure 1.3 Architecture of a DBMS

The DBMS accepts SQL commands generated from a variety of user interfaces, pro duces query evaluation plans, executes these plans against the database, and returns the answers. (This is a simplification: SQL commands can be embedded in host language application programs, e.g., Java or COBOL programs. We ignore these issues to concentrate on the core DBMS functionality.)

When a user issues a query, the parsed query is presented to a query optimizer, which uses information about how the data is stored to produce an efficient execution plan for evaluating the query. An execution plan is a blueprint for evaluating a query, and is usually represented as a tree of relational operators (with annotations that contain additional detailed information about which access methods to use, etc.). We discuss query optimization in Chapter 13. Relational operators serve as the building blocks for evaluating queries posed against the data. The implementation of these operators is discussed in Chapter 12.

The code that implements relational operators sits on top of the file and access methods layer. This layer includes a variety of software for supporting the concept of a file, which, in a DBMS, is a collection of pages or a collection of records. This layer typically supports a heap file, or file of unordered pages, as well as indexes. In addition to keeping track of the pages in a file, this layer organizes the information within a page. File and page level storage issues are considered in Chapter 7. File organizations and indexes are considered in Chapter 8.

The files and access methods layer code sits on top of the buffer manager, which brings pages in from disk to main memory as needed in response to read requests. Buffer management is discussed in Chapter 7.

The lowest layer of the DBMS software deals with management of space on disk, where the data is stored. Higher layers allocate, deallocate, read, and write pages through (routines provided by) this layer, called the disk space manager. This layer is discussed in Chapter 7.

The DBMS supports concurrency and crash recovery by carefully scheduling user re quests and maintaining a log of all changes to the database. DBMS components associ ated with concurrency control and recovery include the transaction manager, which ensures that transactions request and release locks according to a suitable locking pro tocol and schedules the execution transactions; the lock manager, which keeps track of requests for locks and grants locks on database objects when they become available; and the recovery manager, which is responsible for maintaining a log, and restoring the system to a consistent state after a crash. The disk space manager, buffer manager, and file and access method layers must interact with these components. We discuss concurrency control and recovery in detail in Chapter 18.

1.9 PEOPLE WHO DEAL WITH DATABASES

Quite a variety of people are associated with the creation and use of databases. Obvi ously, there are database implementors, who build DBMS software, and end users who wish to store and use data in a DBMS. Database implementors work for ven dors such as IBM or Oracle. End users come from a diverse and increasing number of fields. As data grows in complexity and volume, and is increasingly

recognized as a major asset, the importance of maintaining it professionally in a DBMS is being widely accepted. Many end users simply use applications written by database applica tion programmers (see below), and so require little technical knowledge about DBMS software. Of course, sophisticated users who make more extensive use of a DBMS, such as writing their own queries, require a deeper understanding of its features.

In addition to end users and implementors, two other classes of people are associated with a DBMS: *application programmers* and *database administrators* (DBAs).

Database application programmers develop packages that facilitate data access for end users, who are usually not computer professionals, using the host or data languages and software tools that DBMS vendors provide. (Such tools include report writers, spreadsheets, statistical packages, etc.) Application programs should ideally access data through the external schema. It is possible to write applications that access data at a lower level, but such applications would compromise data independence.

A personal database is typically maintained by the individual who owns it and uses it. However, corporate or enterprise-wide databases are typically important enough and complex enough that the task of designing and maintaining the database is entrusted to a professional called the database administrator. The DBA is responsible for many critical tasks:

Design of the conceptual and physical schemas: The DBA is responsible for interacting with the users of the system to understand what data is to be stored in the DBMS and how it is likely to be used. Based on this knowledge, the DBA must design the conceptual schema (decide what relations to store) and the physical schema (decide how to store them). The DBA may also design widely used portions of the external schema, although users will probably augment this schema by creating additional views.

Security and authorization: The DBA is responsible for ensuring that unau thorized data access is not permitted. In general, not everyone should be able to access all the data. In a relational DBMS, users can be granted permission to access only certain views and relations. For example, although you might al low students to find out course enrollments and who teaches a given course, you would not want students to see faculty salaries or each others' grade information.

The DBA can enforce this policy by giving students permission to read only the Courseinfo view.

Data availability and recovery from failures: The DBA must take steps to ensure that if the system fails, users can continue to access as much of the

uncorrupted data as possible. The DBA must also work to restore the data to a consistent state. The DBMS provides software support for these functions, but the DBA is responsible for implementing procedures to back up the data periodically and to maintain logs of system activity (to facilitate recovery from a crash).

Database tuning: The needs of users are likely to evolve with time. The DBA is responsible for modifying the database, in particular the conceptual and physical schemas, to ensure adequate performance as user requirements change.

1.10 POINTS TO REVIEW

A database management system (DBMS) is software that supports management of large collections of data. A DBMS provides efficient data access, data in dependence, data integrity, security, quick application development, support for concurrent access, and recovery from system failures. (Section 1.1)

Storing data in a DBMS versus storing it in operating system files has many advantages. (Section 1.3)

Using a DBMS provides the user with data independence, efficient data access, automatic data integrity, and security. (Section 1.4)

The structure of the data is described in terms of a *data model* and the description is called a *schema*. The *relational model* is currently the most popular data model. A DBMS distinguishes between *external*, *conceptual*, and *physical schema* and thus allows a view of the data at three levels of abstraction. Physical and logical *data independence*, which are made possible by these three levels of abstraction, insulate the users of a DBMS from the way the data is structured and stored inside a DBMS. (Section 1.5)

A *query language* and a *data manipulation language* enable high-level access and modification of the data. (Section 1.6)

A *transaction* is a logical unit of access to a DBMS. The DBMS ensures that either all or none of a transaction's changes are applied to the database. For performance reasons, the DBMS processes multiple transactions concurrently, but ensures that the result is equivalent to running the transactions one after the other in some order. The DBMS maintains a record of all changes to the data in the *system log*, in order to undo partial transactions and recover from system crashes. *Checkpointing* is a periodic operation that can reduce the time for recovery from a crash. (Section 1.7)

DBMS code is organized into several modules: the disk space manager, the buffer manager, a layer that supports the abstractions of files and index

structures, a layer that implements relational operators, and a layer that optimizes queries and produces an execution plan in terms of relational operators. (Section 1.8)

A *database administrator (DBA)* manages a DBMS for an enterprise. The DBA designs schemas, provide security, restores the system after a failure, and period ically tunes the database to meet changing user needs. Application programmers develop applications that use DBMS functionality to access and manipulate data, and end users invoke these applications. (Section 1.9)

EXERCISES

Exercise 1.1 Why would you choose a database system instead of simply storing data in operating system files? When would it make sense *not* to use a database system?

Exercise 1.2 What is logical data independence and why is it important? Exercise 1.3

Explain the difference between logical and physical data independence.

Exercise 1.4 Explain the difference between external, internal, and conceptual schemas. How are these different schema layers related to the concepts of logical and physical data independence?

Exercise 1.5 What are the responsibilities of a DBA? If we assume that the DBA is never interested in running his or her own queries, does the DBA still need to understand query optimization? Why?

Exercise 1.6 Scrooge McNugget wants to store information (names, addresses, descriptions of embarrassing moments, etc.) about the many ducks on his payroll. Not surprisingly, the volume of data compels him to buy a database system. To save money, he wants to buy one with the fewest possible features, and he plans to run it as a stand-alone application on his PC clone. Of course, Scrooge does not plan to share his list with anyone. Indicate which of the following DBMS features Scrooge should pay for; in each case also indicate why Scrooge should (or should not) pay for that feature in the system he buys.

1. A security facility.
2. Concurrency control.
3. Crash recovery.
4. A view mechanism.
5. A query language.

Exercise 1.7 Which of the following plays an important role in *representing* information about the real world in a database? Explain briefly.

1. The data definition language.

2. The data manipulation language.
3. The buffer manager.

4. The data model.

Exercise 1.8 Describe the structure of a DBMS. If your operating system is upgraded to support some new functions on OS files (e.g., the ability to force some sequence of bytes to disk), which layer(s) of the DBMS would you have to rewrite in order to take advantage of these new functions?

Exercise 1.9 Answer the following questions:

1. What is a transaction?
2. Why does a DBMS interleave the actions of different transactions, instead of executing transactions one after the other?
3. What must a user guarantee with respect to a transaction and database consistency? What should a DBMS guarantee with respect to concurrent execution of several trans actions and database consistency?
4. Explain the strict two-phase locking protocol.
5. What is the WAL property, and why is it important?

PROJECT-BASED EXERCISES

Exercise 1.10 Use a Web browser to look at the HTML documentation for Minibase. Try to get a feel for the overall architecture.

BIBLIOGRAPHIC NOTES

The evolution of database management systems is traced in [248]. The use of data models for describing real-world data is discussed in [361], and [363] contains a taxonomy of data models. The three levels of abstraction were introduced in [155, 623]. The network data model is described in [155], and [680] discusses several commercial systems based on this model. [634] contains a good annotated collection of systems-oriented papers on database management.

Other texts covering database management systems include [169, 208, 289, 600, 499, 656, 669]. [169] provides a detailed discussion of the relational model from a conceptual standpoint and is notable for its extensive annotated bibliography. [499] presents a performance-oriented per spective, with references to several commercial systems. [208] and [600] offer broad coverage of database system concepts, including a discussion of the hierarchical and network data models. [289] emphasizes the connection between database query languages and logic programming. [669] emphasizes data models. Of these texts, [656] provides the most detailed discussion of theoretical issues. Texts devoted to theoretical aspects include [38, 436, 3]. Handbook [653] includes a section on databases that contains introductory survey articles on a number of topics.

2 ENTITY-RELATIONSHIP MODEL

> The great successful men of the world have used their imaginations. They think ahead and create their mental picture, and then go to work materializing that picture in all its details, filling in here, adding a little there, altering this bit and that bit, but steadily building, steadily building.
>
> —Robert Collier

The *entity-relationship (ER) data model* allows us to describe the data involved in a real-world enterprise in terms of objects and their relationships and is widely used to develop an initial database design. In this chapter, we introduce the ER model and discuss how its features allow us to model a wide range of data faithfully.

The ER model is important primarily for its role in database design. It provides useful concepts that allow us to move from an informal description of what users want from their database to a more detailed, and precise, description that can be implemented in a DBMS. We begin with an overview of database design in Section 2.1 in order to motivate our discussion of the ER model. Within the larger context of the overall design process, the ER model is used in a phase called *conceptual database design*.We then introduce the ER model in Sections 2.2, 2.3, and 2.4. In Section 2.5, we discuss database design issues involving the ER model. We conclude with a brief discussion of conceptual database design for large enterprises.

We note that many variations of ER diagrams are in use, and no widely accepted standards prevail. The presentation in this chapter is representative of the family of ER models and includes a selection of the most popular features.

2.1 OVERVIEW OF DATABASE DESIGN

The database design process can be divided into six steps. The ER model is most relevant to the first three steps:

(1) Requirements Analysis: The very first step in designing a database application is to understand what data is to be stored in the database, what applications must be built on top of it, and what operations are most frequent and subject to performance requirements. In other words, we must find out what the users want from the database.

Database design tools: Design tools are available from RDBMS vendors as well as third-party vendors. Sybase and Oracle, in particular, have comprehensive sets design and analysis tools. See the following URL for details on Sybase's tools: http://www.sybase.com/products/application tools The following provides details on Oracle's tools: http://www.oracle.com/tools

This is usually an informal process that involves discussions with user groups, a study of the current operating environment and how it is expected to change, analysis of any available documentation on existing applications that are expected to be replaced or complemented by the database, and so on. Several methodologies have been pro posed for organizing and presenting the information gathered in this step, and some automated tools have been developed to support this process.

(2) Conceptual Database Design: The information gathered in the requirements analysis step is used to develop a high-level description of the data to be stored in the database, along with the constraints that are known to hold over this data. This step is often carried out using the ER model, or a similar high-level data model, and is discussed in the rest of this chapter.

(3) Logical Database Design: We must choose a DBMS to implement our database design, and convert the conceptual database design into a database schema in the data model of the chosen DBMS. We will only consider relational DBMSs, and therefore, the task in the logical design step is to convert an ER schema into a relational database schema. We discuss this step in detail in Chapter 3; the result is a conceptual schema, sometimes called the logical schema, in the relational data model.

2.1.1 Beyond the ER Model

ER modeling is sometimes regarded as a complete approach to designing a logical database schema. This is incorrect because the ER diagram is just an approximate description of the data, constructed through a very subjective evaluation of the infor mation collected during requirements analysis. A more careful analysis can often refine the logical schema obtained at the end of Step 3. Once we have a good logical schema, we must consider performance criteria and design the physical schema. Finally, we must address security issues and ensure that users are able to access the data they need, but not data that we wish to hide from them. The remaining three steps of database design are briefly described below: [1]

[1]This material can be omitted on a first reading of this chapter without loss of continuity.

(4) Schema Refinement: The fourth step in database design is to analyze the collection of relations in our relational database schema to identify potential

problems, and to refine it. In contrast to the requirements analysis and conceptual design steps, which are essentially subjective, schema refinement can be guided by some elegant and powerful theory. We discuss the theory of *normalizing* relations—restructuring them to ensure some desirable properties—in Chapter 15.

(5) Physical Database Design: In this step we must consider typical expected workloads that our database must support and further refine the database design to ensure that it meets desired performance criteria. This step may simply involve build ing indexes on some tables and clustering some tables, or it may involve a substantial redesign of parts of the database schema obtained from the earlier design steps. We discuss physical design and database tuning in Chapter 16.

(6) Security Design: In this step, we identify different user groups and different roles played by various users (e.g., the development team for a product, the customer support representatives, the product manager). For each role and user group, we must identify the parts of the database that they must be able to access and the parts of the database that they should *not* be allowed to access, and take steps to ensure that they can access only the necessary parts. A DBMS provides several mechanisms to assist in this step, and we discuss this in Chapter 17.

In general, our division of the design process into steps should be seen as a classification of the *kinds* of steps involved in design. Realistically, although we might begin with the six step process outlined here, a complete database design will probably require a subsequent tuning phase in which all six kinds of design steps are interleaved and repeated until the design is satisfactory. Further, we have omitted the important steps of implementing the database design, and designing and implementing the application layers that run on top of the DBMS. In practice, of course, these additional steps can lead to a rethinking of the basic database design.

The concepts and techniques that underlie a relational DBMS are clearly useful to someone who wants to implement or maintain the internals of a database system. However, it is important to recognize that serious users and DBAs must also know how a DBMS works. A good understanding of database system internals is essential for a user who wishes to take full advantage of a DBMS and design a good database; this is especially true of physical design and database tuning.

2.2 ENTITIES, ATTRIBUTES, AND ENTITY SETS

An entity is an object in the real world that is distinguishable from other objects. Examples include the following: the Green Dragonzord toy, the toy department, the manager of the toy department, the home address of the manager of the toy depart-

ment. It is often useful to identify a collection of similar entities. Such a collection is called an entity set. Note that entity sets need not be disjoint; the collection of toy department employees and the collection of appliance department employees may

both contain employee John Doe (who happens to work in both departments). We could also define an entity set called Employees that contains both the toy and appliance department employee sets.

An entity is described using a set of attributes. All entities in a given entity set have the same attributes; this is essentially what we mean by *similar*. (This statement is an oversimplification, as we will see when we discuss inheritance hierarchies in Section 2.4.4, but it suffices for now and highlights the main idea.) Our choice of attributes reflects the level of detail at which we wish to represent information about entities. For example, the Employees entity set could use name, social security number (ssn), and parking lot (lot) as attributes. In this case we will store the name, social secu rity number, and lot number for each employee. However, we will not store, say, an employee's address (or gender or age).

For each attribute associated with an entity set, we must identify a domain of possible values. For example, the domain associated with the attribute *name* of Employees might be the set of 20-character strings.[2] As another example, if the company rates employees on a scale of 1 to 10 and stores ratings in a field called *rating*, the associated domain consists of integers 1 through 10. Further, for each entity set, we choose a *key*. A key is a minimal set of attributes whose values uniquely identify an entity in the set. There could be more than one candidate key; if so, we designate one of them as the primary key. For now we will assume that each entity set contains at least one set of attributes that uniquely identifies an entity in the entity set; that is, the set of attributes contains a key. We will revisit this point in Section 2.4.3.

The Employees entity set with attributes *ssn, name,* and *lot* is shown in Figure 2.1. An entity set is represented by a rectangle, and an attribute is represented by an oval. Each attribute in the primary key is underlined. The domain information could be listed along with the attribute name, but we omit this to keep the figures compact. The key is *ssn*.

2.3 RELATIONSHIPS AND RELATIONSHIP SETS

A relationship is an association among two or more entities. For example, we may have the relationship that Attishoo works in the pharmacy department. As with entities, we may wish to collect a set of similar relationships into a relationship set.

[2]To avoid confusion, we will assume that attribute names do not repeat across entity sets. This is not a real limitation because we can always use the entity set name to resolve ambiguities if the same attribute name is used in more than one entity set.

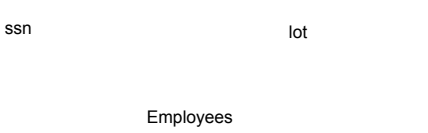

Figure 2.1 The Employees Entity Set

A relationship set can be thought of as a set of *n*-tuples:

$$\{(e_1,...,e_n) \mid e_1 \in E_1,...,e_n \in E_n\}$$

Each *n*-tuple denotes a relationship involving *n* entities e_1 through e_n, where entity e_i is in entity set E_i. In Figure 2.2 we show the relationship set Works In, in which each relationship indicates a department in which an employee works. Note that several relationship sets might involve the same entity sets. For example, we could also have a Manages relationship set involving Employees and Departments.

Figure 2.2 The Works In Relationship Set

A relationship can also have descriptive attributes. Descriptive attributes are used to record information about the relationship, rather than about any one of the par ticipating entities; for example, we may wish to record that Attishoo works in the pharmacy department as of January 1991. This information is captured in Figure 2.2 by adding an attribute, *since*, to Works In. A relationship must be uniquely identified by the participating entities, without reference to the descriptive attributes. In the Works In relationship set, for example, each Works In relationship must be uniquely identified by the combination of employee *ssn* and department *did*. Thus, for a given employee-department pair, we cannot have more than one associated *since* value.

An instance of a relationship set is a set of relationships. Intuitively, an instance can be thought of as a 'snapshot' of the relationship set at some instant in time. An instance of the Works In relationship set is shown in Figure 2.3. Each Employees entity is denoted by its *ssn*, and each Departments entity is denoted by its *did*, for simplicity.

The *since* value is shown beside each relationship. (The 'many-to-many' and 'total participation' comments in the figure will be discussed later, when we discuss integrity constraints.)

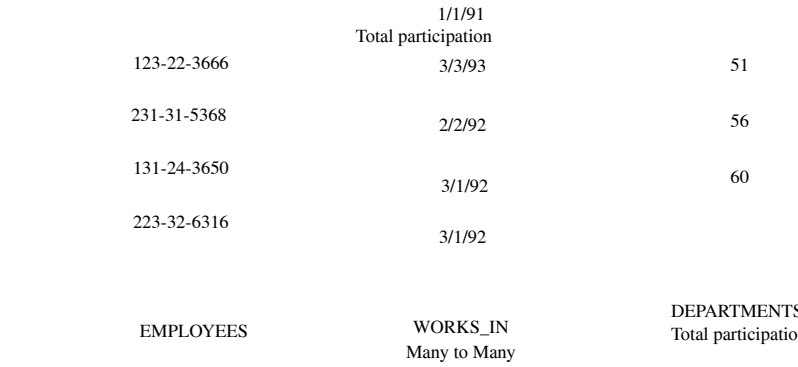

Figure 2.3 An Instance of the Works In Relationship Set

As another example of an ER diagram, suppose that each department has offices in several locations and we want to record the locations at which each employee works. This relationship is ternary because we must record an association between an em ployee, a department, and a location. The ER diagram for this variant of Works In, which we call Works In2, is shown in Figure 2.4.

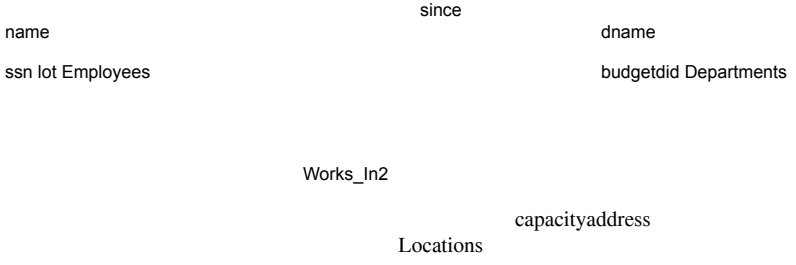

Figure 2.4 A Ternary Relationship Set

The entity sets that participate in a relationship set need not be distinct; sometimes a relationship might involve two entities in the same entity set. For example, consider the Reports To relationship set that is shown in Figure 2.5. Since employees report to other employees, every relationship in Reports To is of the form (emp_1, emp_2),

where both emp_1 and emp_2 are entities in Employees. However, they play different roles: emp_1 reports to the managing employee emp_2, which is reflected in the role indicators *supervisor* and *subordinate* in Figure 2.5. If an entity set plays more than one role, the role indicator concatenated with an attribute name from the entity set gives us a unique name for each attribute in the relationship set. For example, the Reports To relationship set has attributes corresponding to the *ssn* of the supervisor

and the *ssn* of the subordinate, and the names of these attributes are *supervisor ssn* and *subordinate ssn*.

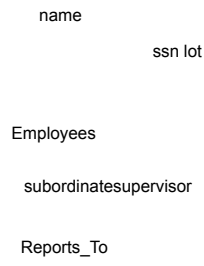

Figure 2.5 The Reports To Relationship Set

2.4 ADDITIONAL FEATURES OF THE ER MODEL

We now look at some of the constructs in the ER model that allow us to describe some subtle properties of the data. The expressiveness of the ER model is a big reason for its widespread use.

2.4.1 Key Constraints

Consider the Works In relationship shown in Figure 2.2. An employee can work in several departments, and a department can have several employees, as illustrated in the Works In instance shown in Figure 2.3. Employee 231-31-5368 has worked in Department 51 since 3/3/93 and in Department 56 since 2/2/92. Department 51 has two employees.

Now consider another relationship set called Manages between the Employees and De partments entity sets such that each department has at most one manager, although a single employee is allowed to manage more than one department. The restriction that each department has at most one manager is an example of a key constraint, and it implies that each Departments entity appears in at most one Manages relationship

in any allowable instance of Manages. This restriction is indicated in the ER diagram of Figure 2.6 by using an arrow from Departments to Manages. Intuitively, the ar row states that given a Departments entity, we can uniquely determine the Manages relationship in which it appears.

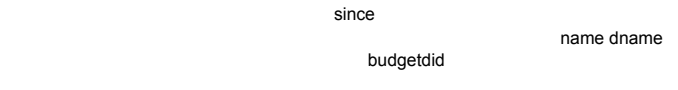

Figure 2.6 Key Constraint on Manages

An instance of the Manages relationship set is shown in Figure 2.7. While this is also a potential instance for the Works In relationship set, the instance of Works In shown in Figure 2.3 violates the key constraint on Manages.

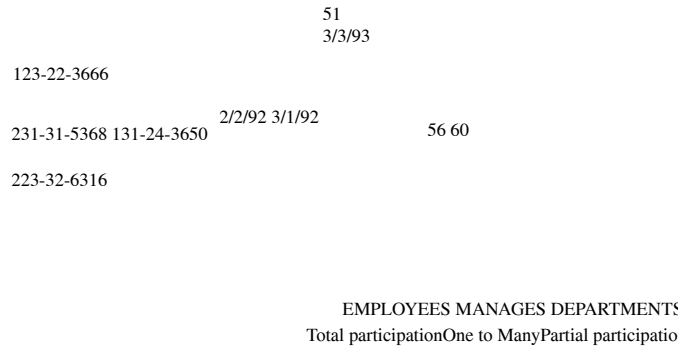

Figure 2.7 An Instance of the Manages Relationship Set

A relationship set like Manages is sometimes said to be one-to-many, to indicate that *one* employee can be associated with *many* departments (in the capacity of a manager), whereas each department can be associated with at most one employee as its manager. In contrast, the Works In relationship set, in which an employee is allowed to work in several departments and a department is allowed to have several employees, is said to be many-to-many.

If we add the restriction that each employee can manage at most one department to the Manages relationship set, which would be indicated by adding an arrow from Employees to Manages in Figure 2.6, we have a one-to-one relationship set.

Key Constraints for Ternary Relationships

We can extend this convention—and the underlying key constraint concept—to rela tionship sets involving three or more entity sets: If an entity set E has a key constraint in a relationship set R, each entity in an instance of E appears in at most one rela tionship in (a corresponding instance of) R. To indicate a key constraint on entity set E in relationship set R, we draw an arrow from E to R.

In Figure 2.8, we show a ternary relationship with key constraints. Each employee

works in at most one department, and at a single location. An instance of the Works In3 relationship set is shown in Figure 2.9. Notice that each department can be associated with several employees and locations, and each location can be associated with several departments and employees; however, each employee is associated with a single department and location.

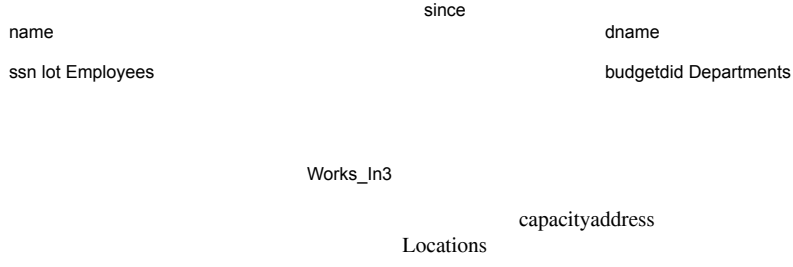

Figure 2.8 A Ternary Relationship Set with Key Constraints

2.4.2 Participation Constraints

The key constraint on Manages tells us that a department has at most one manager. A natural question to ask is whether every department has a manager. Let us say that every department is required to have a manager. This requirement is an example of a participation constraint; the participation of the entity set Departments in the relationship set Manages is said to be total. A participation that is not total is said to be partial. As an example, the participation of the entity set Employees in Manages is partial, since not every employee gets to manage a department.

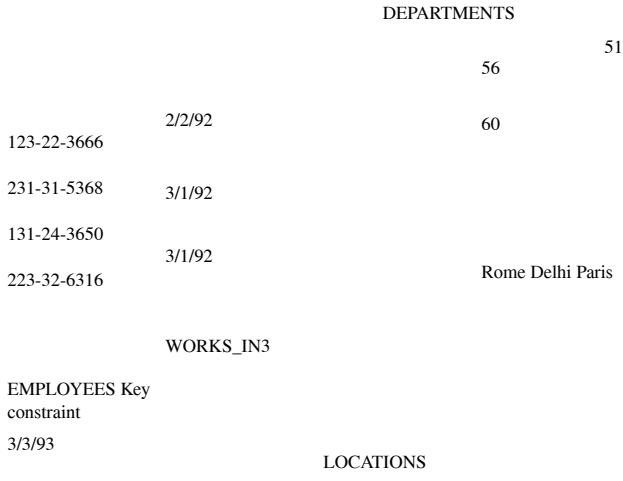

Figure 2.9 An Instance of Works In3

Revisiting the Works In relationship set, it is natural to expect that each employee works in at least one department and that each department has at least one employee. This means that the participation of both Employees and Departments in Works In is total. The ER diagram in Figure 2.10 shows both the Manages and Works In relationship sets and all the given constraints. If the participation of an entity set in a relationship set is total, the two are connected by a thick line; independently, the presence of an arrow indicates a key constraint. The instances of Works In and Manages shown in Figures 2.3 and 2.7 satisfy all the constraints in Figure 2.10.

2.4.3 Weak Entities

Thus far, we have assumed that the attributes associated with an entity set include a key. This assumption does not always hold. For example, suppose that employees can purchase insurance policies to cover their dependents. We wish to record information about policies, including who is covered by each policy, but this information is really our only interest in the dependents of an employee. If an employee quits, any policy owned by the employee is terminated and we want to delete all the relevant policy and dependent information from the database.

We might choose to identify a dependent by name alone in this situation, since it is rea sonable to expect that the dependents of a given employee have different names. Thus the attributes of the Dependents entity set might be *pname* and *age*. The attribute *pname* does *not* identify a dependent uniquely. Recall that the key for Employees is

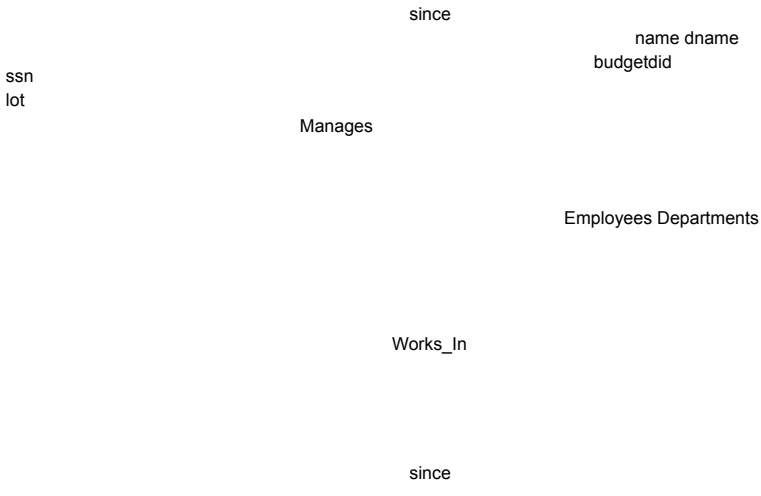

Figure 2.10 Manages and Works In

ssn; thus we might have two employees called Smethurst, and each might have a son called Joe.

Dependents is an example of a weak entity set. A weak entity can be identified uniquely only by considering some of its attributes in conjunction with the primary key of another entity, which is called the identifying owner.

The following restrictions must hold:

- The owner entity set and the weak entity set must participate in a one-to-many relationship set (one owner entity is associated with one or more weak entities, but each weak entity has a single owner). This relationship set is called the identifying relationship set of the weak entity set.
- The weak entity set must have total participation in the identifying relationship set.

For example, a Dependents entity can be identified uniquely only if we take the key of the *owning* Employees entity and the *pname* of the Dependents entity. The set of attributes of a weak entity set that uniquely identify a weak entity for a given owner entity is called a *partial key* of the weak entity set. In our example *pname* is a partial key for Dependents.

The Dependents weak entity set and its relationship to Employees is shown in Fig ure 2.11. The total participation of Dependents in Policy is indicated by linking them

with a dark line. The arrow from Dependents to Policy indicates that each Dependents entity appears in at most one (indeed, exactly one, because of the participation con straint) Policy relationship. To underscore the fact that Dependents is a weak entity and Policy is its identifying relationship, we draw both with dark lines. To indicate that *pname* is a partial key for Dependents, we underline it using a broken line. This means that there may well be two dependents with the same *pname* value.

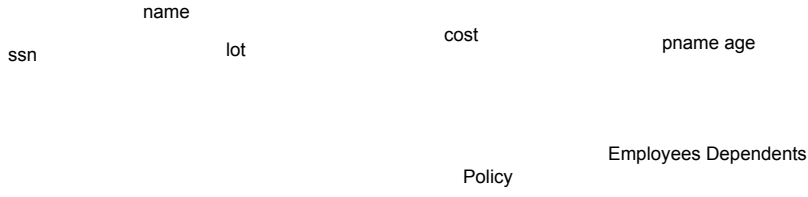

Figure 2.11 A Weak Entity Set

2.4.4 Class Hierarchies

Sometimes it is natural to classify the entities in an entity set into subclasses. For example, we might want to talk about an Hourly Emps entity set and a Contract Emps entity set to distinguish the basis on which they are paid. We might have attributes *hours worked* and *hourly wage* defined for Hourly Emps and an attribute *contractid* defined for Contract Emps.

We want the semantics that every entity in one of these sets is also an Employees entity, and as such must have all of the attributes of Employees defined. Thus, the attributes defined for an Hourly Emps entity are the attributes for Employees plus Hourly Emps. We say that the attributes for the entity set Employees are inherited by the entity set Hourly Emps, and that Hourly Emps ISA (read *is a*) Employees. In addition— and in contrast to class hierarchies in programming languages such as C++—there is a constraint on queries over instances of these entity sets: A query that asks for all Employees entities must consider all Hourly Emps and Contract Emps entities as well. Figure 2.12 illustrates the class hierarchy.

The entity set Employees may also be classified using a different criterion. For example, we might identify a subset of employees as Senior Emps. We can modify Figure 2.12 to reflect this change by adding a second ISA node as a child of Employees and making Senior Emps a child of this node. Each of these entity sets might be classified further, creating a multilevel ISA hierarchy.

A class hierarchy can be viewed in one of two ways:

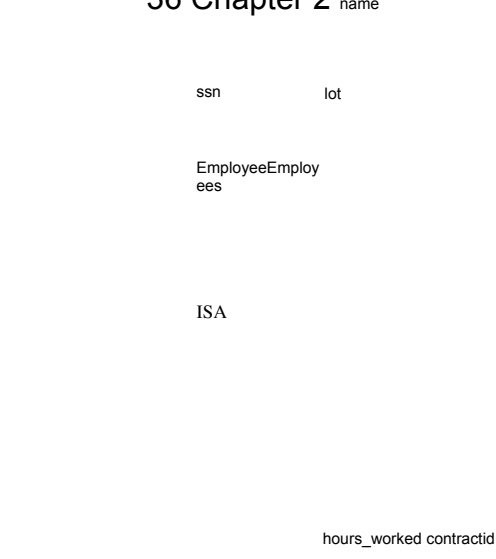

Figure 2.12 Class Hierarchy

Employees is specialized into subclasses. Specialization is the process of iden tifying subsets of an entity set (the superclass) that share some distinguishing characteristic. Typically the superclass is defined first, the subclasses are defined next, and subclass-specific attributes and relationship sets are then added.

Hourly Emps and Contract Emps are generalized by Employees. As another example, two entity sets Motorboats and Cars may be generalized into an entity set Motor Vehicles. Generalization consists of identifying some common charac teristics of a collection of entity sets and creating a new entity set that contains entities possessing these common characteristics. Typically the subclasses are de fined first, the superclass is defined next, and any relationship sets that involve the superclass are then defined.

We can specify two kinds of constraints with respect to ISA hierarchies, namely, *overlap* and *covering* constraints. Overlap constraints determine whether two subclasses are allowed to contain the same entity. For example, can Attishoo be both an Hourly Emps entity and a Contract Emps entity? Intuitively, no. Can he be both a Contract Emps entity and a Senior Emps entity? Intuitively, yes. We denote this by writing 'Con tract Emps OVERLAPS Senior Emps.' In the absence of such a statement, we assume by default that entity sets are constrained to have no overlap.

Covering constraints determine whether the entities in the subclasses collectively include all entities in the superclass. For example, does every Employees entity have to belong to one of its subclasses? Intuitively, no. Does every Motor Vehicles entity have to be either a Motorboats entity or a Cars entity? Intuitively, yes; a charac teristic property of generalization hierarchies is that every instance of a superclass is an instance of a subclass. We denote this by writing 'Motorboats AND Cars COVER

Motor Vehicles.' In the absence of such a statement, we assume by default that there is no covering constraint; we can have motor vehicles that are not motorboats or cars.

There are two basic reasons for identifying subclasses (by specialization or generaliza tion):

1. We might want to add descriptive attributes that make sense only for the entities in a subclass. For example, *hourly wages* does not make sense for a Contract Emps entity, whose pay is determined by an individual contract.

2. We might want to identify the set of entities that participate in some relation ship. For example, we might wish to define the Manages relationship so that the participating entity sets are Senior Emps and Departments, to ensure that only senior employees can be managers. As another example, Motorboats and Cars may have different descriptive attributes (say, tonnage and number of doors),

but as Motor Vehicles entities, they must be licensed. The licensing information can be captured by a Licensed To relationship between Motor Vehicles and an entity set called Owners.

2.4.5 Aggregation

As we have defined it thus far, a relationship set is an association between entity sets. Sometimes we have to model a relationship between a collection of entities and *rela tionships*. Suppose that we have an entity set called Projects and that each Projects entity is sponsored by one or more departments. The Sponsors relationship set cap tures this information. A department that sponsors a project might assign employees to monitor the sponsorship. Intuitively, Monitors should be a relationship set that associates a Sponsors relationship (rather than a Projects or Departments entity) with an Employees entity. However, we have defined relationships to associate two or more *entities*.

In order to define a relationship set such as Monitors, we introduce a new feature of the ER model, called *aggregation*. Aggregation allows us to indicate that a relationship set (identified through a dashed box) participates in another relationship set. This is illustrated in Figure 2.13, with a dashed box around Sponsors (and its participating entity sets) used to denote aggregation. This effectively allows us to treat Sponsors as an entity set for purposes of defining the Monitors relationship set.

When should we use aggregation? Intuitively, we use it when we need to express a relationship among relationships. But can't we express relationships involving other relationships without using aggregation? In our example, why not make Sponsors a ternary relationship? The answer is that there are really two distinct relationships, Sponsors and Monitors, each possibly with attributes of its own. For instance, the

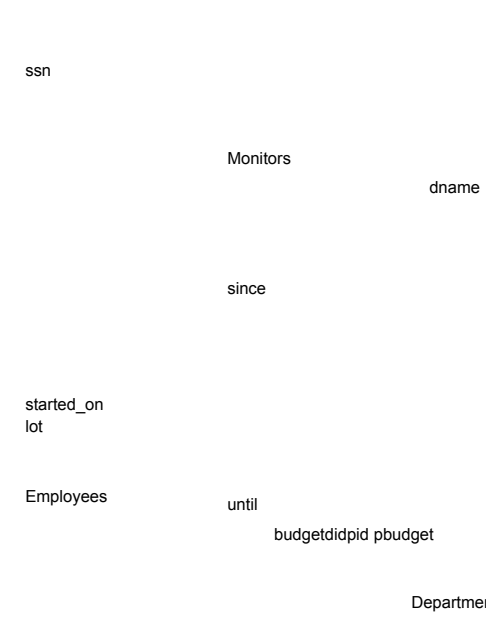

Figure 2.13 Aggregation

Monitors relationship has an attribute *until* that records the date until when the em ployee is appointed as the sponsorship monitor. Compare this attribute with the attribute *since* of Sponsors, which is the date when the sponsorship took effect. The use of aggregation versus a ternary relationship may also be guided by certain integrity constraints, as explained in Section 2.5.4.

2.5 CONCEPTUAL DATABASE DESIGN WITH THE ER MODEL

Developing an ER diagram presents several choices, including the following:

Should a concept be modeled as an entity or an attribute?

Should a concept be modeled as an entity or a relationship?

What are the relationship sets and their participating entity sets? Should we use binary or ternary relationships?

Should we use aggregation?

We now discuss the issues involved in making these choices.

5.

2. 1 Entity versus Attribute

While identifying the attributes of an entity set, it is sometimes not clear whether a property should be modeled as an attribute or as an entity set (and related to the first entity set using a relationship set). For example, consider adding address information to the Employees entity set. One option is to use an attribute *address*. This option is appropriate if we need to record only one address per employee, and it suffices to think of an address as a string. An alternative is to create an entity set called Addresses and to record associations between employees and addresses using a relationship (say, Has Address). This more complex alternative is necessary in two situations:

We have to record more than one address for an employee.

We want to capture the structure of an address in our ER diagram. For example, we might break down an address into city, state, country, and Zip code, in addition to a string for street information. By representing an address as an entity with these attributes, we can support queries such as “Find all employees with an address in Madison, WI.”

For another example of when to model a concept as an entity set rather than as an attribute, consider the relationship set (called Works In2) shown in Figure 2.14.

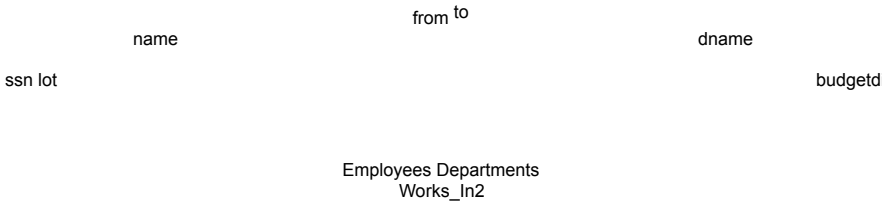

Figure 2.14 The Works In2 Relationship Set

It differs from the Works In relationship set of Figure 2.2 only in that it has attributes *from* and *to*, instead of *since*. Intuitively, it records the interval during which an employee works for a department. Now suppose that it is possible for an employee to work in a given department over more than one period.

This possibility is ruled out by the ER diagram's semantics. The problem is that we want to record several values for the descriptive attributes for each instance of the Works In2 relationship. (This situation is analogous to wanting to record several addresses for each employee.) We can address this problem by introducing an entity set called, say, Duration, with attributes *from* and *to*, as shown in Figure 2.15.

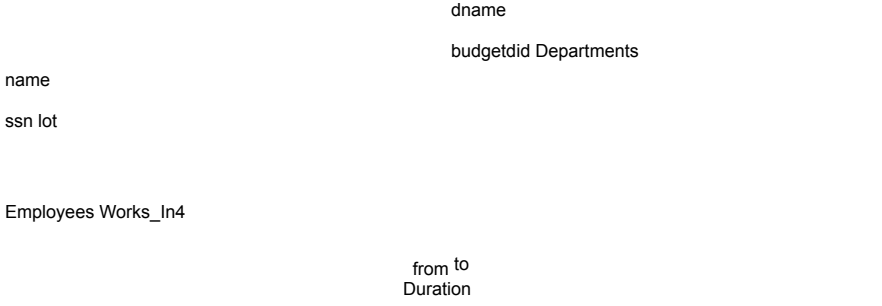

Figure 2.15 The Works In4 Relationship Set

In some versions of the ER model, attributes are allowed to take on sets as values. Given this feature, we could make Duration an attribute of Works In, rather than an entity set; associated with each Works In relationship, we would have a set of intervals. This approach is perhaps more intuitive than modeling Duration as an entity set. Nonetheless, when such set-valued attributes are translated into the relational model, which does not support set-valued attributes, the resulting relational schema is very similar to what we get by regarding Duration as an entity set.

2.5.2 Entity versus Relationship

Consider the relationship set called Manages in Figure 2.6. Suppose that each depart ment manager is given a discretionary budget (*dbudget*), as shown in Figure 2.16, in which we have also renamed the relationship set to Manages2.

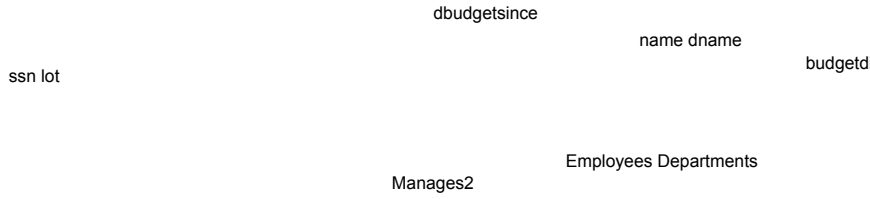

Figure 2.16 Entity versus Relationship

There is at most one employee managing a department, but a given employee could manage several departments; we store the starting date and discretionary budget for each manager-department pair. This approach is natural if we assume that a manager receives a separate discretionary budget for each department that he or she manages.

But what if the discretionary budget is a sum that covers *all* departments managed by that employee? In this case each Manages2 relationship that involves a given employee will have the same value in the *dbudget* field. In general such redundancy could be significant and could cause a variety of problems. (We discuss redundancy and its attendant problems in Chapter 15.) Another problem with this design is that it is misleading.

We can address these problems by associating *dbudget* with the appointment of the employee as manager of a *group* of departments. In this approach, we model the appointment as an entity set, say Mgr Appt, and use a ternary relationship, say Man ages3, to relate a manager, an appointment, and a department. The details of an appointment (such as the discretionary budget) are not repeated for each department that is included in the appointment now, although there is still one Manages3 relation ship instance per such department. Further, note that each department has at most one manager, as before, because of the key constraint. This approach is illustrated in Figure 2.17.

budgetdid

since

Manages3 Mgr_Appts

Departments

dbudget

Figure 2.17 Entity Set versus Relationship

2.5.3 Binary versus Ternary Relationships *

Consider the ER diagram shown in Figure 2.18. It models a situation in which an employee can own several policies, each policy can be owned by several employees, and each dependent can be covered by several policies.

Suppose that we have the following additional requirements:

A policy cannot be owned jointly by two or more employees.

Every policy must be owned by some employee.

 name

ssn

lot

pname age

Covers

Employees Dependents

Policies

policyid cost

Figure 2.18 Policies as an Entity Set

Dependents is a weak entity set, and each dependent entity is uniquely identified by taking *pname* in conjunction with the *policyid* of a policy entity (which, intuitively, covers the given dependent).

The first requirement suggests that we impose a key constraint on Policies with respect to Covers, but this constraint has the unintended side effect that a policy can cover only one dependent. The second requirement suggests that we impose a total participation constraint on Policies. This solution is acceptable if each policy covers at least one dependent. The third requirement forces us to introduce an identifying relationship that is binary (in our version of ER diagrams, although there are versions in which this is not the case).

Even ignoring the third point above, the best way to model this situation is to use two binary relationships, as shown in Figure 2.19.

This example really had two relationships involving Policies, and our attempt to use a single ternary relationship (Figure 2.18) was inappropriate. There are situations, however, where a relationship inherently associates more than two entities. We have seen such an example in Figure 2.4 and also Figures 2.15 and 2.17.

As a good example of a ternary relationship, consider entity sets Parts, Suppliers, and Departments, and a relationship set Contracts (with descriptive attribute *qty*) that involves all of them. A contract specifies that a supplier will supply (some quantity of) a part to a department. This relationship cannot be adequately captured by a collection of binary relationships (without the use of aggregation). With binary relationships, we can denote that a supplier 'can supply' certain parts, that a department 'needs' some

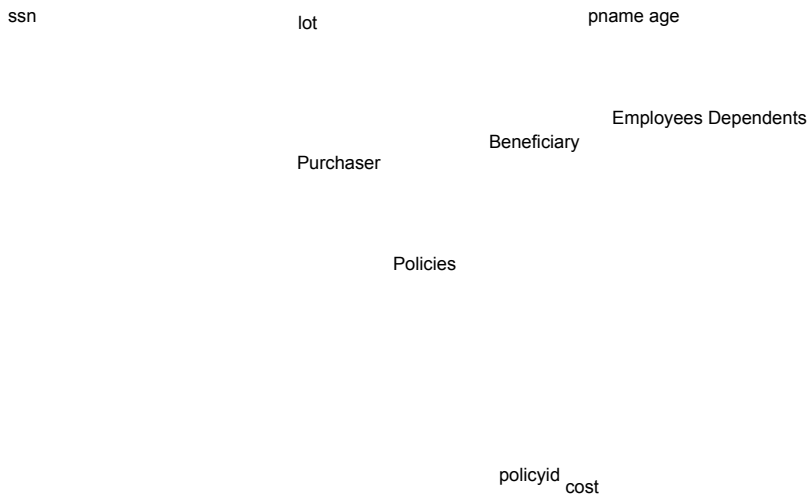

Figure 2.19 Policy Revisited

parts, or that a department 'deals with' a certain supplier. No combination of these relationships expresses the meaning of a contract adequately, for at least two

reasons:

The facts that supplier S can supply part P, that department D needs part P, and that D will buy from S do not necessarily imply that department D indeed buys part P from supplier S!

We cannot represent the *qty* attribute of a contract cleanly.

2.5.4 Aggregation versus Ternary Relationships *

As we noted in Section 2.4.5, the choice between using aggregation or a ternary relation ship is mainly determined by the existence of a relationship that relates a *relationship set* to an entity set (or second relationship set). The choice may also be guided by certain integrity constraints that we want to express. For example, consider the ER diagram shown in Figure 2.13. According to this diagram, a project can be sponsored by any number of departments, a department can sponsor one or more projects, and each sponsorship is monitored by one or more employees. If we don't need to record the *until* attribute of Monitors, then we might reasonably use a ternary relationship, say, Sponsors2, as shown in Figure 2.20.

Consider the constraint that each sponsorship (of a project by a department) be mon itored by at most one employee. We cannot express this constraint in terms of the Sponsors2 relationship set. On the other hand, we can easily express the constraint by drawing an arrow from the aggregated relationship Sponsors to the relationship

 name

ssn

started_on Employees lot dname

budgetdidpid pbudget

DepartmentsProjects Sponsors2

Figure 2.20 Using a Ternary Relationship instead of Aggregation

Monitors in Figure 2.13. Thus, the presence of such a constraint serves as another reason for using aggregation rather than a ternary relationship set.

2.6 CONCEPTUAL DESIGN FOR LARGE ENTERPRISES *

We have thus far concentrated on the constructs available in the ER model for describ ing various application concepts and relationships. The process of conceptual design consists of more than just describing small fragments of the application in terms of ER diagrams. For a large enterprise, the design may require the efforts of more than one designer and span data and application code used by a number of user groups. Using a high-level, semantic data model such as ER diagrams for conceptual design in such an environment offers the additional advantage that the high-level design can be diagrammatically represented and is easily understood by the many people who must provide input to the design process.

An important aspect of the design process is the methodology used to structure the development of the overall design and to ensure that the design takes into account all user requirements and is consistent. The usual approach is that the requirements of various user groups are considered, any conflicting requirements are somehow resolved, and a single set of global requirements is generated at the end of the requirements analysis phase. Generating a single set of global requirements is a difficult task, but it allows the conceptual design phase to proceed with the development of a logical schema that spans all the data and applications throughout the enterprise.

An alternative approach is to develop separate conceptual schemas for different user groups and to then *integrate* these conceptual schemas. To integrate multiple concep tual schemas, we must establish correspondences between entities, relationships, and attributes, and we must resolve numerous kinds of conflicts (e.g., naming conflicts, domain mismatches, differences in measurement units). This task is difficult in its own right. In some situations schema integration cannot be avoided—for example, when one organization merges with another, existing databases may have to be inte grated. Schema integration is also increasing in importance as users demand access to *heterogeneous* data sources, often maintained by different organizations.

2.7 POINTS TO REVIEW

Database design has six steps: requirements analysis, conceptual database design, logical database design, schema refinement, physical database design, and security design. Conceptual design should produce a high-level description of the data, and the entity-relationship (ER) data model provides a graphical approach to this design phase. (Section 2.1)

In the ER model, a real-world object is represented as an *entity*. An *entity set* is a collection of structurally identical entities. Entities are described using *attributes*. Each entity set has a distinguished set of attributes called a *key* that

can be used to uniquely identify each entity. (Section 2.2)

A *relationship* is an association between two or more entities. A *relationship set* is a collection of relationships that relate entities from the same entity sets. A relationship can also have *descriptive attributes*. (Section 2.3)

A *key constraint* between an entity set S and a relationship set restricts instances of the relationship set by requiring that each entity of S participate in at most one relationship. A *participation constraint* between an entity set S and a relationship set restricts instances of the relationship set by requiring that each entity of S participate in at least one relationship. The identity and existence of a *weak entity* depends on the identity and existence of another (*owner*) entity. *Class hierarchies* organize structurally similar entities through inheritance into sub- and super classes. *Aggregation* conceptually transforms a relationship set into an entity set such that the resulting construct can be related to other entity sets. (Section 2.4)

Development of an ER diagram involves important modeling decisions. A thor ough understanding of the problem being modeled is necessary to decide whether to use an attribute or an entity set, an entity or a relationship set, a binary or ternary relationship, or aggregation. (Section 2.5)

Conceptual design for large enterprises is especially challenging because data from many sources, managed by many groups, is involved. (Section 2.6)

EXERCISES

Exercise 2.1 Explain the following terms briefly: *attribute, domain, entity, relationship, entity set, relationship set, one-to-many relationship, many-to-many relationship, participa tion constraint, overlap constraint, covering constraint, weak entity set, aggregation*, and *role indicator*.

Exercise 2.2 A university database contains information about professors (identified by so cial security number, or SSN) and courses (identified by courseid). Professors teach courses; each of the following situations concerns the Teaches relationship set. For each situation, draw an ER diagram that describes it (assuming that no further constraints hold).

1. Professors can teach the same course in several semesters, and each offering must be recorded.
2. Professors can teach the same course in several semesters, and only the most recent such offering needs to be recorded. (Assume this condition applies in all subsequent questions.)
3. Every professor must teach some course.
4. Every professor teaches exactly one course (no more, no less).
5. Every professor teaches exactly one course (no more, no less), and every course must be taught by some professor.
6. Now suppose that certain courses can be taught by a team of professors jointly, but it is possible that no one professor in a team can teach the course. Model this situation,

introducing additional entity sets and relationship sets if necessary.

Exercise 2.3 Consider the following information about a university database:

Professors have an SSN, a name, an age, a rank, and a research specialty.

Projects have a project number, a sponsor name (e.g., NSF), a starting date, an ending date, and a budget.

Graduate students have an SSN, a name, an age, and a degree program (e.g., M.S. or Ph.D.).

Each project is managed by one professor (known as the project's principal investigator).

Each project is worked on by one or more professors (known as the project's co-investigators). Professors can manage and/or work on multiple projects.

Each project is worked on by one or more graduate students (known as the project's research assistants).

When graduate students work on a project, a professor must supervise their work on the project. Graduate students can work on multiple projects, in which case they will have a (potentially different) supervisor for each one.

Departments have a department number, a department name, and a main office.

Departments have a professor (known as the chairman) who runs the department.

Professors work in one or more departments, and for each department that they work in, a time percentage is associated with their job.

Graduate students have one major department in which they are working on their degree.

Each graduate student has another, more senior graduate student (known as a student advisor) who advises him or her on what courses to take.

Design and draw an ER diagram that captures the information about the university. Use only the basic ER model here, that is, entities, relationships, and attributes. Be sure to indicate any key and participation constraints.

Exercise 2.4 A company database needs to store information about employees (identified by *ssn*, with *salary* and *phone* as attributes); departments (identified by *dno*, with *dname* and *budget* as attributes); and children of employees (with *name* and *age* as attributes). Employees *work* in departments; each department is *managed by* an employee; a child must be identified uniquely by *name* when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.

Draw an ER diagram that captures this information.

Exercise 2.5 Notown Records has decided to store information about musicians who perform on its albums (as well as other company data) in a database. The company has wisely chosen to hire you as a database designer (at your usual consulting fee of $2,500/day).

Each musician that records at Notown has an SSN, a name, an address, and a phone number. Poorly paid musicians often share the same address, and no address has more than one phone.

Each instrument that is used in songs recorded at Notown has a name (e.g., guitar,

synthesizer, flute) and a musical key (e.g., C, B-flat, E-flat).

Each album that is recorded on the Notown label has a title, a copyright date, a format (e.g., CD or MC), and an album identifier.

Each song recorded at Notown has a title and an author.

Each musician may play several instruments, and a given instrument may be played by several musicians.

Each album has a number of songs on it, but no song may appear on more than one album.

Each song is performed by one or more musicians, and a musician may perform a number of songs.

Each album has exactly one musician who acts as its producer. A musician may produce several albums, of course.

Design a conceptual schema for Notown and draw an ER diagram for your schema. The following information describes the situation that the Notown database must model. Be sure to indicate all key and cardinality constraints and any assumptions that you make. Identify any constraints that you are unable to capture in the ER diagram and briefly explain why you could not express them.

Exercise 2.6 Computer Sciences Department frequent fliers have been complaining to Dane County Airport officials about the poor organization at the airport. As a result, the officials have decided that all information related to the airport should be organized using a DBMS, and you've been hired to design the database. Your first task is to organize the informa tion about all the airplanes that are stationed and maintained at the airport. The relevant information is as follows:

Every airplane has a registration number, and each airplane is of a specific model.

The airport accommodates a number of airplane models, and each model is identified by a model number (e.g., DC-10) and has a capacity and a weight.

A number of technicians work at the airport. You need to store the name, SSN, address, phone number, and salary of each technician.

Each technician is an expert on one or more plane model(s), and his or her expertise may overlap with that of other technicians. This information about technicians must also be recorded.

Traffic controllers must have an annual medical examination. For each traffic controller, you must store the date of the most recent exam.

All airport employees (including technicians) belong to a union. You must store the union membership number of each employee. You can assume that each employee is uniquely identified by the social security number.

The airport has a number of tests that are used periodically to ensure that airplanes are still airworthy. Each test has a Federal Aviation Administration (FAA) test number, a name, and a maximum possible score.

The FAA requires the airport to keep track of each time that a given airplane is tested by a given technician using a given test. For each testing event, the information needed is the date, the number of hours the technician spent doing the test, and the score that the airplane received on the test.

1. Draw an ER diagram for the airport database. Be sure to indicate the various attributes of each entity and relationship set; also specify the key and participation constraints for each relationship set. Specify any necessary overlap and covering constraints as well (in English).
2. The FAA passes a regulation that tests on a plane must be conducted by a technician who is an expert on that model. How would you express this constraint in the ER diagram? If you cannot express it, explain briefly.

Exercise 2.7 The Prescriptions-R-X chain of pharmacies has offered to give you a free life time supply of medicines if you design its database. Given the rising cost of health care, you agree. Here's the information that you gather:

Patients are identified by an SSN, and their names, addresses, and ages must be recorded.

Doctors are identified by an SSN. For each doctor, the name, specialty, and years of experience must be recorded.

Each pharmaceutical company is identified by name and has a phone number.

For each drug, the trade name and formula must be recorded. Each drug is sold by a given pharmaceutical company, and the trade name identifies a drug uniquely from among the products of that company. If a pharmaceutical company is deleted, you need not keep track of its products any longer.

Each pharmacy has a name, address, and phone number.

Every patient has a primary physician. Every doctor has at least one patient.

Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another.

Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and a quantity associated with it. You can assume that if a doctor prescribes the same drug for the same patient more than once, only the last such prescription needs to be stored.

Pharmaceutical companies have long-term contracts with pharmacies. A pharmaceutical company can contract with several pharmacies, and a pharmacy can contract with several pharmaceutical companies. For each contract, you have to store a start date, an end date, and the text of the contract.

Pharmacies appoint a supervisor for each contract. There must always be a supervisor for each contract, but the contract supervisor can change over the lifetime of the contract.

1. Draw an ER diagram that captures the above information. Identify any constraints that are not captured by the ER diagram.
2. How would your design change if each drug must be sold at a fixed price by all pharma cies?
3. How would your design change if the design requirements change as follows: If a doctor prescribes the same drug for the same patient more than once, several such prescriptions may have to be stored.

Exercise 2.8 Although you always wanted to be an artist, you ended up being an expert on databases because you love to cook data and you somehow confused 'data base' with 'data baste.' Your old love is still there, however, so you set up a database company, ArtBase, that builds a product for art galleries. The core of this product is a database with a schema that captures all the information that galleries need to maintain. Galleries keep information about artists, their names (which are unique), birthplaces, age, and style of art. For each piece of artwork, the artist, the year it was made, its unique title, its type of art (e.g., painting, lithograph, sculpture, photograph), and its price must be stored. Pieces of artwork are also classified into groups of various kinds, for example, portraits, still lifes, works by Picasso, or works of the 19th century; a given piece may belong to more than one group. Each group is identified by a name (like those above) that describes the group. Finally, galleries keep information about customers. For each customer, galleries keep their unique name, address, total amount of dollars they have spent in the gallery (very important!), and the artists and groups of art that each customer tends to like.

Draw the ER diagram for the database.

BIBLIOGRAPHIC NOTES

Several books provide a good treatment of conceptual design; these include [52] (which also contains a survey of commercial database design tools) and [641].

The ER model was proposed by Chen [145], and extensions have been proposed in a number of subsequent papers. Generalization and aggregation were introduced in [604]. [330] and [514] contain good surveys of semantic data models. Dynamic and temporal aspects of semantic data models are discussed in [658].

[642] discusses a design methodology based on developing an ER diagram and then translating to the relational model. Markowitz considers referential integrity in the context of ER to relational mapping and discusses the support provided in some commercial systems (as of that date) in [446, 447].

The entity-relationship conference proceedings contain numerous papers on conceptual design, with an emphasis on the ER model, for example, [609].

View integration is discussed in several papers, including [84, 118, 153, 207, 465, 480, 479, 596, 608, 657]. [53] is a survey of several integration approaches.

3 THE RELATIONAL MODEL

TABLE: An arrangement of words, numbers, or signs, or combinations of them, as in parallel columns, to exhibit a set of facts or relations in a definite, compact, and comprehensive form; a synopsis or scheme.

—Webster's *Dictionary of the English Language*

Codd proposed the relational data model in 1970. At that time most database systems were based on one of two older data models (the hierarchical model and the network model); the relational model revolutionized the database field and largely supplanted these earlier models. Prototype relational database management systems were devel oped in pioneering research projects at IBM and UC-Berkeley by the mid-70s, and several vendors were offering relational database products shortly thereafter. Today, the relational model is by far the dominant data model and is the foundation for the leading DBMS products, including IBM's DB2 family, Informix, Oracle, Sybase, Mi crosoft's Access and SQLServer, FoxBase, and Paradox. Relational database systems are ubiquitous in the marketplace and represent a multibillion dollar industry.

The relational model is very simple and elegant; a database is a collection of one or more *relations*, where each relation is a table with rows and columns. This simple tabular representation enables even novice users to understand the contents of a database, and it permits the use of simple, high-level languages to query the data. The major advantages of the relational model over the older data models are its simple data representation and the ease with which even complex queries can be expressed.

This chapter introduces the relational model and covers the following issues:

How is data represented?

What kinds of integrity constraints can be expressed?

How can data be created and modified?

How can data be manipulated and queried?

How do we obtain a database design in the relational model?

How are logical and physical data independence achieved?

SQL: It was the query language of the pioneering System-R relational DBMS developed at IBM. Over the years, SQL has become the most widely used language for creating, manipulating, and querying relational DBMSs. Since many vendors offer SQL products, there is a need for a standard that defines 'official SQL.' The existence of a standard allows users to measure a given vendor's version of SQL for completeness. It also allows users to distinguish SQL features that are specific to one product from those that are standard; an application that relies on non-standard features is less portable.

The first SQL standard was developed in 1986 by the American National Stan dards Institute (ANSI), and was called SQL-86. There was a minor revision in 1989 called SQL-89, and a major revision in 1992 called SQL-92. The Interna tional Standards Organization (ISO) collaborated with ANSI to develop SQL-92. Most commercial DBMSs currently support SQL-92. An exciting development is the imminent approval of SQL:1999, a major extension of SQL-92. While the cov erage of SQL in this book is based upon SQL-92, we will cover the main extensions of SQL:1999 as well.

While we concentrate on the underlying concepts, we also introduce the Data Def inition Language (DDL) features of SQL-92, the standard language for creating, manipulating, and querying data in a relational DBMS. This allows us to ground the discussion firmly in terms of real database systems.

We discuss the concept of a relation in Section 3.1 and show how to create relations using the SQL language. An important component of a data model is the set of constructs it provides for specifying conditions that must be satisfied by the data. Such conditions, called *integrity constraints* (ICs), enable the DBMS to reject operations that might corrupt the data. We present integrity constraints in the relational model in Section 3.2, along with a discussion of SQL support for ICs. We discuss how a DBMS enforces integrity constraints in Section 3.3. In Section 3.4 we turn to the mechanism for accessing and retrieving data from the database, *query languages*, and introduce the querying features of SQL, which we examine in greater detail in a later chapter.

We then discuss the step of converting an ER diagram into a relational database schema in Section 3.5. Finally, we introduce *views*, or tables defined using queries, in Section 3.6. Views can be used to define the external schema for a database and thus provide the support for logical data independence in the relational model.

3.1 INTRODUCTION TO THE RELATIONAL MODEL

The main construct for representing data in the relational model is a relation.A relation consists of a relation schema and a relation instance. The relation instance

is a table, and the relation schema describes the column heads for the table. We first describe the relation schema and then the relation instance. The schema specifies the relation's name, the name of each field (or column, or attribute), and the domain of each field. A domain is referred to in a relation schema by the domain name and has a set of associated values.

We use the example of student information in a university database from Chapter 1 to illustrate the parts of a relation schema:

Students(*sid:* string, *name:* string, *login:* string, *age:* integer, *gpa:* real)

This says, for instance, that the field named *sid* has a domain named string. The set of values associated with domain string is the set of all character strings.

We now turn to the instances of a relation. An instance of a relation is a set of tuples, also called records, in which each tuple has the same number of fields as the relation schema. A relation instance can be thought of as a *table* in which each tuple is a *row*, and all rows have the same number of fields. (The term *relation instance* is often abbreviated to just *relation*, when there is no confusion with other aspects of a relation such as its schema.)

An instance of the Students relation appears in Figure 3.1. The instance *S*1 contains

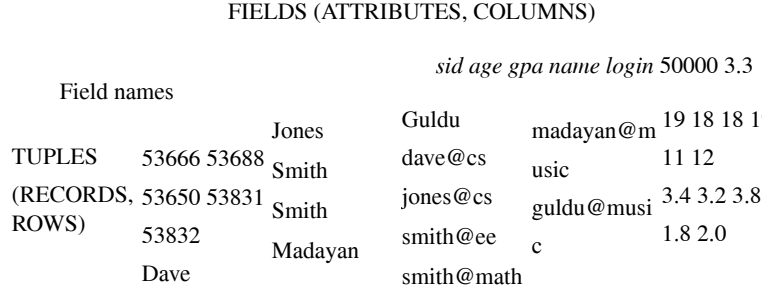

Figure 3.1 An Instance *S*1 of the Students Relation

six tuples and has, as we expect from the schema, five fields. Note that no two rows are identical. This is a requirement of the relational model—each relation is defined to be a *set* of unique tuples or rows.[1] The order in which the rows are listed is not important. Figure 3.2 shows the same relation instance. If the fields are named, as in

[1]In practice, commercial systems allow tables to have duplicate rows, but we will assume that a relation is indeed a set of tuples unless otherwise noted.

sid	*name*	*login*	*age*	*gpa*

53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0
53688	Smith	smith@ee	18	3.2
53650	Smith	smith@math	19	3.8
53666	Jones	jones@cs	18	3.4
50000	Dave	dave@cs	19	3.3

Figure 3.2 An Alternative Representation of Instance *S*1 of Students

our schema definitions and figures depicting relation instances, the order of fields does not matter either. However, an alternative convention is to list fields in a specific order and to refer to a field by its position. Thus *sid* is field 1 of Students, *login* is field 3, and so on. If this convention is used, the order of fields is significant. Most database systems use a combination of these conventions. For example, in SQL the named fields convention is used in statements that retrieve tuples, and the ordered fields convention is commonly used when inserting tuples.

A relation schema specifies the domain of each field or column in the relation instance. These domain constraints in the schema specify an important condition that we want each instance of the relation to satisfy: The values that appear in a column must be drawn from the domain associated with that column. Thus, the domain of a field is essentially the *type* of that field, in programming language terms, and restricts the values that can appear in the field.

More formally, let R(f_1:D1, ..., f_n:Dn) be a relation schema, and for each f_i,1 ≤ i ≤ n, let Dom_i be the set of values associated with the domain named Di. An instance of R that satisfies the domain constraints in the schema is a set of tuples with n fields:

$$\{hf_1 : d_1, \ldots, f_n : d_n i \mid d_1 \in Dom_1, \ldots, d_n \in Dom_n\}$$

The angular brackets *h...i* identify the fields of a tuple. Using this notation, the first Students tuple shown in Figure 3.1 is written as *hsid:* 50000, *name:* Dave, *login:* dave@cs, *age:* 19, *gpa:* 3.3*i*. The curly brackets *{...}* denote a set (of tuples, in this definition). The vertical bar | should be read 'such that,' the symbol ∈ should be read 'in,' and the expression to the right of the vertical bar is a condition that must be satisfied by the field values of each tuple in the set. Thus, an instance of R is defined

as a set of tuples. The fields of each tuple must correspond to the fields in the relation schema.

Domain constraints are so fundamental in the relational model that we will henceforth consider only relation instances that satisfy them; therefore, *relation instance* means *relation instance that satisfies the domain constraints in the relation schema.*

The degree, also called arity, of a relation is the number of fields. The cardinality of a relation instance is the number of tuples in it. In Figure 3.1, the degree of the relation (the number of columns) is five, and the cardinality of this instance is six.

A relational database is a collection of relations with distinct relation names. The relational database schema is the collection of schemas for the relations in the database. For example, in Chapter 1, we discussed a university database with rela tions called Students, Faculty, Courses, Rooms, Enrolled, Teaches, and Meets In. An instance of a relational database is a collection of relation instances, one per rela tion schema in the database schema; of course, each relation instance must satisfy the domain constraints in its schema.

3.1.1 Creating and Modifying Relations Using SQL-92

The SQL-92 language standard uses the word *table* to denote *relation*, and we will often follow this convention when discussing SQL. The subset of SQL that supports the creation, deletion, and modification of tables is called the Data Definition Lan guage (DDL). Further, while there is a command that lets users define new domains, analogous to type definition commands in a programming language, we postpone a dis cussion of domain definition until Section 5.11. For now, we will just consider domains that are built-in types, such as integer.

The CREATE TABLE statement is used to define a new table.[2] To create the Students relation, we can use the following statement:

```
CREATE TABLE Students ( sid CHAR(20),
                        name CHAR(30),
                        login CHAR(20),
                        age INTEGER,
                        gpa REAL )
```

Tuples are inserted using the INSERT command. We can insert a single tuple into the Students table as follows:

```
INSERT
INTO Students (sid, name, login, age, gpa)
VALUES (53688, 'Smith', 'smith@ee', 18, 3.2)
```

We can optionally omit the list of column names in the INTO clause and list the values in the appropriate order, but it is good style to be explicit about column names.

[2]SQL also provides statements to destroy tables and to change the columns associated with a table; we discuss these in Section 3.7.

We can delete tuples using the DELETE command. We can delete all Students tuples with *name* equal to Smith using the command:

```
DELETE
FROM Students S
WHERE S.name = 'Smith'
```

We can modify the column values in an existing row using the UPDATE command. For example, we can increment the age and decrement the gpa of the student with *sid* 53688:

```
UPDATE Students S
SET S.age = S.age + 1, S.gpa = S.gpa - 1
WHERE S.sid = 53688
```

These examples illustrate some important points. The WHERE clause is applied first and determines which rows are to be modified. The SET clause then determines how these rows are to be modified. If the column that is being modified is also used to determine the new value, the value used in the expression on the right side of equals (=) is the *old* value, that is, before the modification. To illustrate these points further, consider the following variation of the previous query:

```
UPDATE Students S
SET S.gpa = S.gpa - 0.1
WHERE S.gpa >= 3.3
```

If this query is applied on the instance *S*1 of Students shown in Figure 3.1, we obtain the instance shown in Figure 3.3.

sid	*name*	*login*	*age*	*gpa*
50000	Dave	dave@cs	19	3.2
53666	Jones	jones@cs	18	3.3
53688	Smith	smith@ee	18	3.2

53650	Smith	smith@math	19	3.7
53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

Figure 3.3 Students Instance *S*1 after Update

3.2 INTEGRITY CONSTRAINTS OVER RELATIONS

A database is only as good as the information stored in it, and a DBMS must therefore help prevent the entry of incorrect information. An integrity constraint (IC) is a